



ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering
Chandubhai S. Patel Institute of Technology
Department of Mechanical Engineering
Bachelor of Technology Programme
(Third Year Mechanical Engineering)





CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Mechanical Engineering) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
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www.charusat.ac.in

Year 2024-25

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS Bachelor of Technology Programmes Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

- | | |
|-----|---|
| (i) | Undergraduate programme (B.Tech) |
| | Minimum 8 semesters (4 academic years) |
| | Maximum 16 semesters (8 academic years) |

3. Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – 1 attached

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The continuous assessment will be conducted by the respective department/ institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or combination of these.

7.1.3 The weightages of continuous assessment and end-semester University examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examinations

7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations are as follows:

Minimum percentage marks to be obtained in end-semester University examination (for applicable courses)	Minimum overall percentage marks to be obtained in each course
40%	45%

7.2.2 If a candidate obtains minimum required percentage of marks in end-semester University examination in an applicable course but fails to obtain minimum required overall percentage of marks, he/she has to repeat the examination till the minimum required overall percentage of marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.5$ & ≤ 10.0
First class	$CGPA \geq 6.0$ & < 7.5
Second Class	$CGPA \geq 5.0$ & < 6.0
Pass Class	$CGPA < 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

10. Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM
FOR
BACHELOR OF TECHNOLOGY & ENGINEERING

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 4 types of courses: Foundation courses, Core courses, Elective courses and Non-credit (audit) courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or

which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

D. Cluster Elective Course (CE):

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.4. Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be consider to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5. Medium of Instruction

The Medium of Instruction will be English.

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

The objectives of the Mechanical Engineering Undergraduate Program are to produce graduate who can:

1. Apply their knowledge and expertise to find solutions of the problems in core and allied areas of engineering, exhibiting creativity and entrepreneurial skills.
2. Conduct themselves in an ethical and professional manner, exhibiting effective skills in communication, management, teamwork and leadership.
3. Continue their intellectual development through self-study and continuing education, practicing life-long learning.

Program Outcomes: Mechanical Engineering

- PO1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes: Mechanical Engineering

- PSO1 The Mechanical Engineering graduates will be able to analyze, design, and evaluate the performance of mechanical components and systems by using various tools.
- PSO2 The Mechanical Engineering graduates will be able to plan, automate and evaluate manufacturing and system processes.

TEACHING & EXAMINATION SCHEME
B. TECH. PROGRAMME IN MECHANICAL ENGINEERING

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT) (w.e.f. June 2024)											
TEACHING & EXAMINATION SCHEME FOR THIRD YEAR B. TECH. PROGRAMME IN MECHANICAL ENGINEERING											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
SY Sem-5	ME 350	Thermal Engineering	3	0	3	3	30	70	00	00	100
	ME341	Design Concepts & Machine Drawing	4	2	6	5	30	70	25	25	150
	ME342	Heat & Mass Transfer	4	2	6	5	30	70	25	25	150
	ME352	Metrology & Quality Control	3	2	5	4	30	70	25	25	150
	ME3XX	Programme Elective - I	3	2	5	4	30	70	25	25	150
	ME349	Summer Internship - I	-			3	00	00	75	75	150
	HSI31.02A/ HSI42 A/ HSI43 A/ HSI44 A	Communication and Soft Skills/ French/ German/ Spanish	2		2	2	00	00	30	70	100
			27			26					950
SY Sem-6	ME344	Refrigeration and Air Conditioning	3	2	5	4	30	70	25	25	150
	ME348	Production Technology	3	2	5	4	30	70	25	25	150
	ME354	Fluid Machines	3	2	5	4	30	70	25	25	150
	ME355	Design of Machine Elements	3	0	3	3	30	70	00	00	100
	ME353	Computer Aided Design	3	2	5	4	30	70	25	25	150
	ME356	Python for Mechanical Engineers	0	0	2	1	00	00	25	25	50
	HSI32.02 A	Contributory Personality Development	2		2	2	00	00	30	70	100
	ME3XX	Programme Elective - II	3	0	3	3	30	70	00	00	100
			30			25					950

Programme Electives:

Sr. No.	Course Nature	Course Code	Course Title
1	Programme Elective-I	ME373	Metal Casting Technology
2		ME377	Dynamics of Compressible Flow
3		ME378	Noise, Vibration & Harshness
4		ME379	Hydraulics & Pneumatics
5		ME380	Combustion Processes
6	Programme Elective-II	ME381.01	Renewable Energy Sources
7		ME382.01	Industrial Engineering & Management
8		ME383.01	Digital Manufacturing
9		ME384.01	Instrumentation & Control
10		ME385.01	Automobile Engineering

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME 350 THERMAL ENGINEERING

5th Semester and 3rd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Vapour Power Cycles	12
2.	Gas Power Cycles	09
3.	Fuels and Combustion	09
4.	Heat Engine and Internal Combustion Engine	11
5.	Cryogenic Systems	04

Total hours (Theory): 45

Total: 75

B. Detailed Syllabus:

- | | | |
|--------------|---|----------------------------|
| 1 | Vapour Power Cycles | 12 Hours 27% |
| 1.1 | Carnot Vapour Power Cycle, Rankine Cycle | |
| 1.2 | Mean Temperature of Heat Addition, Capacity of Steam Power Plant | |
| 1.3 | Modified Rankine Cycle, Comparison of Rankine and Carnot Cycles. | |
| 1.4 | Deviation of Actual Cycle from Ideal Cycle. | |
| 1.5 | Reheat Cycle, Ideal Regenerative Cycles, Reheat–Regenerative Cycle. | |
| 1.6 | Regenerative Feed heating cycle, Open Feed Water heater, Closed Feed Water Heater, Multi Stage Regenerative Cycles. | |
| 1.7 | Carnotization of Rankine Cycle, Optimum Degree of Regeneration, Binary Vapour Power Cycle, Mercury Steam Binary Vapour Power Cycle. | |
| 1.8 | Thermodynamics of Coupled Cycles, Process heat and By–Product Power (back pressure turbine), Exergy Analysis of Vapour Power Cycle. | |
|
2 |
Gas Power Cycles |
09 Hours 20% |

- 2.1 Introduction, Carnot Cycle, Stirling Cycle (Regenerative Cycle), Ericsson Cycle
- 2.2 Air Standard Otto Cycle, Air Standard Diesel Cycle
- 2.3 Limited Pressure Cycle or Dual Cycle, Comparison of Otto, Diesel and Dual Cycles
- 2.4 Atkinson Cycle, Lenoir Cycle, Brayton Cycle (Simple Gas Turbine Cycle, Brayton – Rankine Combine Cycle)

3 Fuels and Combustion

09 Hours 20%

- 3.1 Introduction, Classification of Fuels, Solid, Liquid and Gaseous Fuels, Merits and Demerits of Solid, Liquid and Gaseous Fuels
- 3.2 Requirements of a good fuel, Chemical Structure of Petroleum, Properties required for SI and CI engine fuels, Rating of IC engine fuels
- 3.3 Biofuels: Classification, Bioethanol – Merits, Demerits, Methods of producing Bioethanol, Biodiesel – Merits, Demerits, Methods of producing Biodiesel
- 3.4 Jatropha – an important Biodiesel Plant, Calorific Value of Fuels ,Higher Calorific Value, Lower Calorific Value, Experimental determination of HCV , Bomb Calorimeter, Boy's Gas Calorimeter
- 3.5 Combustion Equations of Solid Fuels, Combustion Equations of Gaseous Fuels, Theoretical or Minimum Air required for complete combustion of Solid Fuels and Gaseous Fuels
- 3.6 Mass of Carbon in flue gases, Mass of Flue Gases per kg of Fuel, Excess Air Supplied, Flue Gas Analysis by Orsat Apparatus

4 Heat Engine and Internal Combustion Engine

11 Hours 25%

- 4.1 Introduction, Energy Conversion, definition of Engine, definition of heat engine, Elements of Heat Engine, Classification of Heat engine, Comparison between External and Internal Combustion Engine, Classification of IC Engines, Application of IC engine
- 4.2 Engine performance parameters: Bore, Stroke, Top Dead Centre, Bottom Dead Centre, Clearance Volume, Swept Volume, Compression Ratio, Indicated and brake Mean Effective Pressure, Mean Piston Speed, Specific Power Output, Specific Fuel Consumption, Indicated power, Frictional Power, Brake power, Inlet–Valve Mach Index, Fuel–Air (F/A) or Air–Fuel Ratio (A/F) Indicated Thermal Efficiency, Brake Thermal Efficiency, Mechanical Efficiency, Volumetric Efficiency, Relative Efficiency or Efficiency Ratio
- 4.3 Comparison of Air standard and air fuel, Effect of operating variable on Air fuel cycle, difference between Air–Fuel cycle and Actual cycle
- 4.4 Indicator Diagram, Heat balance sheet of IC Engine
- 4.5 Various Systems in IC Engines

- 4.6 SI Engines and CI Engines, Different Parts of IC engines, Valve Timing diagram of IC Engine, Comparison between 2 stroke and 4 stroke engine, Comparison between SI and CI Engine
- 4.7 Supercharging and Turbo charging of IC engine, Objectives of supercharging, Effect of Supercharging on SI and CI engine

5 Cryogenic Systems

04 Hours 08%

- 5.1 Mechanical Properties at low temperatures. Properties of Cryogenic Fluids. Applications of Cryogenic fluids.
- 5.2 Gas Liquefaction

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

CO1: Illustrate the various thermal cycle for power generation.

CO2: Analyze the performance of vapor power cycle and gas power cycle.

CO3: Measure the various properties of fuel.

CO4: Compute the performance of IC engine.

CO5: Explain the concept of cryogenic system.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Nag P. K., "Engineering Thermodynamics", Tata Mc Graw–Hill publications. Fifth edition.
2. Cengel Yunus A, "Thermodynamics an engineering approach", Tata McGRAW Hill Publications. Eighth edition
3. Ganeshan. V, "Internal Combustion Engine", Tata Mc–Graw–Hill. Fourth edition.

Reference Books:

1. Rudramoorthy R, "Thermal Engineering", Tata McGraw–Hill publications
2. Arora and Domkundwar, "Power Plant Engineering", Dhanpat Rai Publications
3. Rajput. R. K. "Internal Combustion Engine", Laxmi Publications
4. Patel. R. C. "Elements of Heat engine" vol 1,2 and 3, Acharya Publications
5. Sarkar. B , "Thermal Engineering", Tata Mc Graw – Hill publications.
6. Rao Y. V. C., "Theory and Problems of Thermodynamics", Wiley Eastern Ltd
7. Klaus, D. T. and Thomas, F. M., "Cryogenic Process Engineering", Plenum Press, 2001.

Web material:

1. www.howstuffworks.com
2. <http://www.wikipedia.org>
3. <http://www.fing.edu.uy/if/mirror/TEST/testhome/indexstates.html>

Other materials

1. Mechanical Engineering (IE)
2. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna
3. Steam Tables
4. Compressibility Chart
5. Proceeding of combustion institute
6. Experimental Thermal and Fluid Science
7. International Journal of Thermal Science

ME – 341 DESIGN CONCEPTS & MACHINE DRAWING

5th Semester and 3rd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Design Consideration and Engineering Materials	03
2.	Concepts of Mechanics of Solids	09
3.	Riveted and Welded Joints	12
4.	Threaded and Bolted Joints	05
5.	Cotter and Knuckle Joints	06
6.	Column And Struts	04
7.	Lever Design	04
8.	Spring Design	04
9.	Production Drawings	07
10.	Computer Aided Drafting	06

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

A Machine Design

- | | | | |
|-----|---|----------|-----|
| 1 | Design Consideration and Engineering Materials | 03 Hours | 05% |
| 1.1 | Definition ,classification and general design procedure | | |
| 1.2 | Classification and Mechanical Properties of materials | | |
| 2 | Concepts of Mechanics of Solids | 09 Hours | 15% |
| 2.1 | Types of Axial, Bending, and Torsional loading and stresses | | |
| 2.2 | Principal stresses and principal planes concept | | |

2.3	Theories of Failures and Selection and use of Theories of Failures		
2.4	Curved Beams and combined loading situations in Machine parts		
3	Riveted and Welded joints	12 Hours	20%
3.1	Methods of riveting, materials, types of rivets		
3.2	Important terms used for rivets, caulking and fullering		
3.3	Design of rivets		
3.4	Eccentric loaded riveted joints		
3.5	Types of weld joints and symbols		
3.6	Strength of welded joints		
3.7	Stresses of welded joints		
3.8	Axially loaded unsymmetrical and eccentric loaded welded joints		
4	Threaded and Bolted joints	05 Hours	08 %
4.1	Types of screw thread and its terminology		
4.2	Stresses in screwed fastening due to static loading		
4.3	Bolted joint under eccentric loading		
5	Cotter and Knuckle Joints	06 Hours.	10 %
5.1	Types of cotter joint and its design		
5.2	Design of cotter foundation bolt		
5.3	Design of knuckle joints		
6	Column and Struts	04 Hours.	07 %
6.1	Compressive axial loading of columns and struts, Slenderness ratio		
6.2	Compressive stress and Buckling of members, Effect of end conditions		
6.3	Euler's Formula, Applications, validity and limitations		
6.4	Rankine's Formula		
7	Lever Design	04 Hours.	07 %
7.1	General Procedure for design of levers		
7.2	Design of lever for safety valve, design of bell crank lever		
7.3	Design of rocker arm for exhaust valves		

8 Spring Design 04 Hours. 07%

- 8.1 Design of spring, buckling, surge in spring
- 8.2 Spring in series and parallel, concentric and composite spring
- 8.3 Design of leaf spring

B Machine Drawing

9 Production drawings 07 Hours. 11 %

- 9.1 Limits, fits and tolerances
- 9.2 Geometric tolerances
- 9.3 Surface texture
- 9.4 Shop floor drawing

10 Computer aided drafting 06 Hours 10 %

- 10.1 Introduction to CAD
- 10.2 Use of CAD software and relevant commands
- 10.3 Drafting of 2-D drawing in Auto-CAD software

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

Upon successful completion of this course, the student will be able to:

CO1: Describe types of engineering design and different properties of engineering materials.

CO2: Compute the stress acting on various machine elements under various failure theories.

CO3: Summarize about temporary and permanent joints based on application requirements.

CO4: Predict the load behavior that can be applied to column, lever and spring

CO5: Use different computer aided drafting tools for modeling of Industrial Drawing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	1	-	-	-	1	-
CO2	2	3	2	1	-	-	-	-	-	-	-	-	2	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-	1	-
CO4	2	2	-	1	-	-	-	-	-	-	-	-	1	-
CO5	3	3	1	2	3	-	-	-	1	-	-	2	2	-

F. Recommended Study Material:

Text Books:

1. Bhandari V B, "Design of Machine Elements", Tata McGraw Hill publishing Co.
2. Pandya N and Shah C, "Machine Design", Charotar Publishers.

Reference books:

1. Sharma P. C., Aggarwal D. K. "Machine Design", S.K. Kataria & Sons, 2009.
2. Bhatt N. D., "Machine Drawing", Charotar Publication.
3. Shigley Joseph Edward, Mischke Charles R., "Mechanical Engineering Design", McGraw Hill International Edition.
4. Hibbler R. C., "Mechanics of Materials", Pearson Education, Inc., 2017.
5. Gere J M and Goodno B J, "Mechanics of Materials", CENGAGE learning
6. Orthein William, "Machine Component Design (Vol. I & II)", M/s. Jaico Publishing.
7. Kulkarni S. G., "Machine Design – Solved Problems", Tata McGraw Hill Publishing Company Ltd., New Delhi
8. Sham Tickoo, "AutoCAD", and CENGAGE learning Indian Edition.

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT%20Kharagpur/Machine%20design1/New_index1.html
2. PSG Design data book

Other materials

Software: Computer Aided Drafting Packages/software like AutoCAD, Pro-Engineer etc.

Journals:

1. Mechanical Engineering (IE)
2. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna
3. Materials and Design, ELSEVIER Journal

ME – 342 HEAT & MASS TRANSFER

5th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Heat Conduction	19
3.	Convective Heat Transfer	12
4.	Thermal Radiation	10
5.	Heat Exchangers	10
6.	Mass Transfer	04
7.	Boiling and Condensation	03

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction | 02 Hours | 05% |
| 1.1 | Basic modes of Heat transfer, Introduction to laws, some heat transfer parameters. | | |
| 2 | Heat Conduction | 19 Hours | 30% |
| 2.1 | Thermal conductivity and other relevant properties. Fourier's law. General three dimensional heat conduction equation in Cartesian, Cylindrical and Spherical co – ordinates. | | |
| 2.2 | One dimensional, heat conduction without and with heat generation through plane slabs, cylinders and spheres. Transient Conduction: The Lumped Capacitance Method | | |
| 2.3 | Concept of thermal resistance, Electrical analogy, Heat transfer through composite slabs, cylinders and spheres, contact resistance. | | |

- 2.4 Critical thickness of insulation for cylinder and sphere.
- 2.5 Heat transfer from finned surfaces, general equation, Efficiency and effectiveness of fins.
Transient conduction, Lumped capacitance model, One dimensional transient problem – analytical solutions, One dimensional Heisler charts.
- 3 Convective Heat Transfer 12 Hours 25%**
- 3.1 Convective Heat Transfer coefficient, Energy equation, Thermal boundary layer, dimensional analysis.
- 3.2 Forced Convection: Blasius solution for laminar boundary layer flows, Van-Karman integral momentum equation, and Energy equation of thermal Boundary layer over a flat plate, Approximate solution of energy equation, Laminar Tube flow, Turbulent boundary layer, Turbulent flow over flat plate and in tubes.
- 3.3 Free Convection: Momentum and Energy equation for laminar flow over flat plate, Integral equation for momentum and energy on a flat plate, Empirical relations and their use.
- 4 Thermal Radiation 10 Hours 20%**
- 4.1 Concept of radiation, surface emission properties, absorptivity, reflectivity and transmissivity, black, white and grey surfaces.
- 4.2 Laws of radiation – Planck, Stefan Boltzman, Wein's displacement, Kirchoff. Intensity of radiation and solid angle, Lambert's cosine law, shape factor.
- 4.3 Radiation exchange between non black bodies, radiation shield, heat exchange between two grey surfaces, electrical analogy.
- 5 Heat Exchangers 10Hours 10%**
- 5.1 Heat exchanger types, flow arrangements, overall heat transfer coefficient, fouling factor, LMTD for parallel flow and counter flow heat exchangers.
- 5.2 Effectiveness– NTU method, expression for effectiveness of a parallel flow and counter flow heat exchangers. Multi-pass and cross flow heat exchangers.
- 6 Mass Transfer 04 Hours 05%**
- 6.1 Analogy between heat and mass transfer, Fick's law of diffusion, Mass transfer coefficient, Evaporation of water into air, Schmidt number, Sherwood number.
- 7 Boiling and Condensation 03 Hours 05%**

7.1 Different regimes of boiling, mechanism of condensation, Nusselt's theory of film condensation on a vertical surface, use of correlations in solving film wise condensation on plane surfaces, horizontal tubes and tube banks.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Use the principles of heat transfer in steady and unsteady state conduction problems.

CO2: Interpret the principles of heat transfer by convection.

CO3: Analyze the heat transfer in extended surfaces.

CO4: Apply the concepts of radiation heat transfer for practical applications.

CO5: Calculate various parameters of heat exchangers using the principles of heat transfer.

CO6: Describe the principles of mass transfer.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	-	-	-	-	-	-	-	-	2	-
CO2	3	2	-	1	-	-	-	-	-	-	-	-	2	-
CO3	3	2	1	2	-	-	-	-	-	-	-	-	2	-
CO4	3	2	-	2	-	-	-	-	-	-	-	-	2	-
CO5	3	3	1	1	-	-	-	-	-	-	-	-	1	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Kumar D. S. “Heat and Mass Transfer” S. K. Kataria and Sons, 2015
2. Holman J. P., “Heat Transfer”, McGraw Hill, 2009.
3. Sukhatme S. P., “Heat Transfer”, Universities Press (India), 2005.

Reference Books:

1. Sachdeva R. C. “Fundamentals of Engineering Heat and Mass transfer” New Age International, 2017.
2. Yunus A Cengel. “Heat & Mass Transfer” McGraw Hill, 2014.
3. E.R.G. Eckert and Robert M. Drake, “Heat and Mass Transfer”, McGraw Hill, 1996.
4. Chapman A. J., “Heat Transfer”, Macmillan, New York, 2000.
5. Kothandraman. C. P., “Fundamentals of Heat and Mass transfer”, New Age International, 2012.

Web Material:

1. <http://www.nptel.iitm.ac.in/>

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com(International journal of Heat and fluid flow)

ME – 352 METROLOGY & QUALITY CONTROL

5th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	General Concepts and Errors in Measurement	05
2.	Linear and Angular Measurements	05
3.	Screw Thread and Gear Measurements	06
4.	Measurement of Surface Finish and Machine Tool Tests	05
5.	Miscellaneous Measurements	03
6.	Modern Measurement Techniques	03
7.	Quality Management	18

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|-----|
| 1 | General Concepts and Errors in Measurement | 05 Hours | 11% |
| 1.1 | General concepts | | |
| 1.2 | Static and Dynamic Characteristics, Calibration, Limits, fits tolerances and allowances, IS codes for limits and fits. | | |
| 1.3 | Errors in Measurement: Static and Dynamic errors. | | |
| 2 | Linear and Angular Measurements | 05 Hours | 11% |

2.1	Introduction and classification of Linear Measuring Instruments (e.g. Vernier Caliper, Micrometer, Height and Depth Gauge, Comparators)		
2.2	Applications, Advantages and Limitations of commonly used linear measuring instruments		
3	Screw Thread and Gear Measurements	06 Hours	13%
3.1	Screw Thread Measurement: Instruments and procedures of Screw Thread Measurement		
3.2	Gear Measurement: Instruments and procedures of Gear Measurement		
4	Measurement of Surface Finish and Machine Tool Tests	05 Hours	11%
4.1	Terminology and Methods to measure surface finish and its elements, Machine Tool Tests		
5	Miscellaneous Measurements & Modern Measurement Techniques	06 Hours	13%
5.1	CMM, UMM, AIS, Machine Vision etc.		
6	Quality Management	18 Hours	40%
6.1	Inspection and Quality Control: Terminology of Inspection and QC		
6.2	Acceptance Sampling: Terminology and Different Plans for Acceptance		
6.3	Quality Assurance (QA): Activities under quality assurance		
6.4	Statistical Process Control (SPC): Techniques accessed for SPC		
6.5	Total Quality Management (TQM): Key Principles and Approaches for TQM		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Clarify the importance of measurement with General concepts with errors, calibration in Mechanical systems.

CO2: Describe importance of precise measurement and role of accuracy in manufacture industries.

CO3: Use of fundamental measuring instruments.

CO4: Adopt different measurements techniques for measuring specific properties like pressure, temperature, surface roughness, velocity, acceleration etc.

CO5: Recognizing the systems of limits, fits and tolerances and correlate with machine drawing and manufacturing processes.

CO6: Analyze and interpret the data obtained from the different measurements processes and present it in the graphical form, statistical form for understanding the concepts of SQC by Constructing and interpreting various control charts.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	-	-	2	2	-
CO2	3	2	2	2	-	3	-	2	-	-	1	2	2	1
CO3	3	2	2	-	-	3	-	-	-	-	-	2	2	1
CO4	3	3	2	2	-	3	-	2	2	1	1	2	2	3
CO5	3	2	2	2	-	3	-	2	2	1	1	2	2	3
CO6	3	3	2	3	2	3	-	2	2	1	2	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Jain R. K., “Engineering Metrology”, Khanna Publisher, 2009.
2. M.Mahajan, “A Textbook of Metrology”, Dhapat Rai & Company, 2nd edition (2019).
2. Doeblein E. O., “Measurement Systems, Application Design”, McGraw Hill, 5th Edition (2004).
3. Kumar D.S., “Mechanical Measurements and Control”, Metropolitan, New Delhi, 4th Edition.
4. L. C. Jhamb “Production (Operation) Management”, Everest Publishing House, 12th edition.
5. M. Mahajan “Statistical Quality Control”, Dhanpat Rai & Co. (P) Ltd, 1997.
6. M. Mahajan, “Industrial Engineering and Production Management”, Dhanpat Rai & Company, 3rd edition

Reference Books:

1. Rajput R.K., “Mechanical Measurements and Instrumentation”, S. K. Kataria and Sons.
2. Sawhney A. K., “A Course in Mechanical Measurements and Instrumentation”.
3. Sawhney A. K. and Mahajan, “Text Book of Measurement and Metrology”, Dhanpat Rai and Co (p) Ltd.
4. Hume K. J., “Engineering Metrology”, Kalyani Publisher (2001).
5. Beckwith T.G , “Mechanical Measurements”, Pearson Publishers, 5th Edition
6. Venkateshan S. P., “Mechanical Measurements”, Ane Books India, (2008).
7. Jalgaonkar R.V., “Mechanical Measurement and Control”, Everest Publishing House, Pune.
8. B. C. Nakra and K. K. Chaudhry, “Instrumentation, Measurement and Analysis”, Tata McGraw Hill Publication, (2004).
9. RegaRajendra, “Principles of Engineering Metrology”, Jaico Publication, 1st Edition (2008).
10. Richard S. Figliola and Donald E. Beasley, “Theory and Design for Mechanical Measurements”, Wiley, 5th Edition (2010).
11. Tayal A. K., “Instrumentation Mechanical Measurement”, Galgotia Publisher, 2nd Edition (2007).
12. Groover M. P., “Automation, Production Systems and Computer Integrated Manufacturing”, Prentice Hall of India Private Limited, 2nd Edition (2007).
13. Bewoor, Vinaykulkarni, “Metrology and Measurement”, Tata McGraw Hill, 1st Edition (2009).
14. Joseph Juran and A. Blanton Godfrey, “Juran's Quality Handbook”, McGraw Hill, 1999
15. K C Jain and A K Chitale “ Quality Assurance and Total Quality Management (ISO 9000, QS 9000 ISO 14000) ” Khanna Publishers, 2003

Web Material:

1. http://nptel.iitm.ac.in/courses/IITMADRAS/Mechanical_Measurements_Metrology/index.php
2. [http://nptel.ac.in/courses/Webcourse-contents/IIT-ROORKEE/INDUSTRIAL ENGINERRING](http://nptel.ac.in/courses/Webcourse-contents/IIT-ROORKEE/INDUSTRIAL_ENGINERRING)
3. Quality forum of India. <http://qcfi.in>.
4. Quality engineering journal. <https://www.tandfonline.com/loi/lqen20>
5. International Journal of Quality Engineering and Technology
<http://www.inderscience.com/jhome.php?jcode=ijqet>
6. American society of quality <https://asq.org.in/>

Other Material:

1. Mechanical Engineering Metrology by M. Adithan, Atlantic Publishers.
2. M. Adithan and R. Bahn, “Metrology Lab”, Manual T.T.T.I. Chandigarh
3. Sadhna: <http://www.ias.ac.in/sadhana/>
4. www.sciencedirect.com (Flow measurement and Instrumentation)

IS/ International Codes:

IS 919 – 1993 Recommendation for limits, fits and tolerances
IS 2029 – 1962 Dial gauges
IS 2909 – 1964 Guide for selection of fits
IS 2921 – 1964 Vernier height gauges
IS 2984 – 1966 Slip gauges
IS 3139 – 1966 Dimensions for screw threads
IS 3179 – 1965 Feeler gauges
IS 3455 – 1966 Tolerances for plain limit gauges
IS 3477 – 1973 Snap gauges
IS 6137 – 1971 Plain plug gauges
IS 3651 – 1976 Vernier Caliper
IS 4440 – 1967 Slip gauges accessories
IS 5359 – 1969 Sine bars
IS 5939 – 1970 Sine angles, sine tables
IS 2949 – 1964 V Block

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester - V

HS13L.02: COMMUNICATION AND SOFT SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

Objectives of the Course:

- To hone and sharpen Communication Skills of students
- To prepare globally and multi-culturally competent communicators and professionally compatible cadre of future professionals
- To equip and empower students to qualify and successfully clear all the phases of selection procedure for on and off campus interviews

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1. An Introduction to Communication 06 Hours 20%

Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication

2. **Cross-cultural Communication and Globalization** 03 Hours 10%
Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication
3. **Communication for Career Building** 10 Hours 33%
Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose
4. **Group Dynamics and Soft Skills** 05 Hours 17%
An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset
5. **Effective Presentation Strategies** 04 Hours 14%
Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion
6. **Contemporary Issues in Communication and Soft Skills** 02 Hours 06%
Trends and Practices in Communication, Case Studies

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

- Text book:

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
 2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
 3. Raman, M & Singh, P. Business Communication, Oxford University Press
- **Reference book:**
 1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
 2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited
 - **Web material:**
 1. <https://www.coursera.org/learn/careerdevelopment>
 2. <https://www.futurelearn.com/courses/writing-applications>
 3. <https://www.futurelearn.com/courses/workplace-englis>

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester V
HSI42 A: French

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - French
- to learn basic grammar and sentence construction of French language
- to learn listening, speaking, reading and writing in French language
- to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to French Language	08
2.	French for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	26.66%
	• Facts and figures of French Language • Basic French Linguistics • * Alphabets * Accents * Liaison * Nasalization • French Culture, Differ between French and English • Grammar: ➤ Subject Pronoun ➤ Verbs: (être, avoir, habiter, regarder, manger ... “er” verb) ➤ Form of address ➤ Greetings ➤ Numbers (1 to 20) ➤ Nouns and plurals of nouns □ The expression: C’est, Il y a • Presentation: ➤ Self-Introduction ➤ Question and answering • Dialogue		
2.	French for Basic Social Conversations	06 Hours	20%
	• Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present tense ➤ Positive Forms ➤ Negative Forms ➤ Numbers (21 to 100, 100-1000) ➤ Days and Months • Social Links: ➤ My family & relations ➤ Gathering information from someone • Dialogue		
3.	Formal Conversations - I	06 Hours	20%

	• Grammar: ➤ Common Adjectives ➤ Interrogative Forms ➤ Directions, Countries, Nationalities, Seasons ➤ Professions ➤ Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “verb) • Work, Study and Travel □ Job/ Profession		
	□ Ticket Reservation (At Bus/At Railway/At Airport) □ Dialogue		
4.	Formal Conversations - II	06 Hours	20%
	• Grammar: ➤ Common Prepositions ➤ Common Conjunctions ➤ Future with verb “Aller” ➤ Colors, Shapes, Animals, Vegetables, Fruits • Food & Shopping ➤ Buy a vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) • Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Comparative Adjectives □ Indiquer le lieu □ The expression: “On” ➤ Irregular Verbs • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of French language and culture
CO2	use French for everyday social conversations
CO3	use basic French to talk in professional settings

CO4	talk in formal situations with basic French
CO5	use French for basic language functions

Evaluation:

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

- ❖ Text book: -
- ❖ Reference Reading: The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with French 1 Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-french-1/contentsection-overview?active-tab=description-tab>
2. Getting started with French 2 Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-french-2/contentsection-overview?active-tab=description-tab>
3. Getting started with French 3 Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-french-3/contentsection-overview?active-tab=description-tab>

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester V
HS143 A: German

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - German
- to learn basic grammar and sentence construction of German language
- to learn listening, speaking, reading and writing in German language □ to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to German Language	08
2.	German for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to German Language	08 Hours	26.66%
	• Facts and figures of German Language • Basic German Linguistics ➤ Alphabets ➤ Accents ➤ Liaison ➤ Nasalization • German Culture, Difference between German and English • Grammar: ➤ Basic Sentence Formation ➤ Personal Pronoun ➤ Basic questions & Answers ➤ Numbers (0 - 20) ➤ Verbs: (Sein, Haben, Werden) ➤ Greetings • Presentation: ➤ Self-Introduction ➤ Question and answering		
2.	German for Basic Social Conversations	06 Hours	20%
	• Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present tense (Positive Forms and Negative Forms) ➤ Numbers (21-10000) ➤ Days, Months, Family Relation ➤ Jahreszeiten ➤ Cloths and Body Parts ➤ Weather • Social Links: Gathering information from someone • Dialogue		
3.	Formal Conversations - I	06 Hours	20%

	□ Grammar: ➤ Verbs: (Arbeiten, Wohnen, Kommen, Machen , Sprechen , Gehen) ➤ Possessive Adjectives ➤ Prepositions ➤ Conjunctions: und, aber, oder, denn, dann ➤ Profession ➤ W – Frage		
	□ Im – Am -Um • Work, Study and Travel □ Ticket Reservation (At Bus/At Railway/At Airport) • Dialogue		
4.	Formal Conversations - II	06 Hours	20%
	• Grammar: ➤ Modal Verbs ➤ Nomminative ➤ Akkusative ➤ Negation: nicht, kein, ➤ Irregular Verbs ➤ Common Conjunctions ➤ Colors • Food & Shopping ➤ Buy vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market • Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Dative ➤ Grocery Items ➤ Mode of Transport ➤ Extra Vocabulary • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of German language and culture
CO2	use German for everyday social conversations
CO3	use basic German to talk in professional settings
CO4	talk in formal situations in basic German language
CO5	use German for basic language functions

Evaluation

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

- ❖ Text book: -
- ❖ Reference Reading: The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with German 1
Platform: The Open University
Link <https://www.open.edu/openlearn/languages/getting-started-german-1/contentsection-overview?active-tab=description-tab>
2. Getting started with German 2
Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-german-2/content-section-overview?active-tab=description-tab>
3. Getting started with German 3
Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-german-3/contentsection-overview?active-tab=description-tab>

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester V
HS144 A: Spanish

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

- --

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - Spanish
- to learn basic grammar and sentence construction of Spanish language
- to learn listening, speaking, reading and writing in Spanish language □ to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Spanish Language	08
2.	Spanish for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Spanish Language	08 Hours	26.66%
	- Facts and figures of Spanish Language - Basic Spanish Linguistics ➤ Alphabets ➤ Accents ➤ Liaison ➤ Nasalization - Spanish Culture, Difference between Spanish and English - Grammar: ➤ Subject Pronouns ➤ Noun – Plural Nouns ➤ Greetings (Formal & Informal) ➤ Numbers (1 to 20) ➤ Present Tense Part -1 ➤ The expression: (There is, There are) ➤ Days, Months and Seasons - Presentation: ➤ Self-Introduction ➤ Question and answering		
2.	Spanish for Basic Social Conversations	06 Hours	20%
	- Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present Tense Part -2 (Positive Forms and Negative Forms) ➤ Numbers (21 to 100) ➤ Regular Verbs ➤ Nation and Nationality ➤ Present Tense Part-3 : (has/have) - Social Links: - Gathering information from someone □ Dialogue		
3.	Formal Conversations - I	06 Hours	20%

	• Grammar: ➤ Common Adjectives ➤ Characteristics ➤ Interrogative Forms ➤ Directions ➤ Professions ➤ Verbs • Work, Study and Travel		
	□ Ticket Reservation (At Bus/At Railway/At Airport) □ Dialogue		
4.	Formal Conversations - II	06 Hours	20%
	• Grammar: ➤ Common Prepositions ➤ Common Conjunctions ➤ Time and Date ➤ Colours ➤ Vegetables and Fruits ➤ Irregular Verbs • Food & Shopping ➤ Buy vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market □ Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Family Relation ➤ Reflexive Verbs ➤ Professions ➤ Animals • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of Spanish language and culture
CO2	use Spanish for everyday social conversations
CO3	talk in formal situations with basic Spanish
CO4	utilise basic Spanish to talk in formal settings

CO5	use Spanish for basic language functions
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Evaluation

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

- ❖ Text book: -
- ❖ Reference Reading: The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with Spanish 1
Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-spanish-1/content-section-overview?active-tab=description-tab>
2. Getting started with Spanish 2
Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-spanish-2/content-section-overview?active-tab=description-tab>
3. Getting started with Spanish 3
Platform: The Open University
Link: <https://www.open.edu/openlearn/languages/getting-started-spanish-3/contentsection-overview?active-tab=description-tab>

ME – 349 SUMMER INTERNSHIP - I

5th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Project	Total	Credit
Hours/week	30	90	3
Marks	150	150	

A. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research center is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this

guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.

- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share

- **INTRODUCTION:** In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

C. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Integrate class room theory with real-world engineering problems of industry/organization

CO2: Develop work habits, attitudes and ethics necessary for industrial job.

CO3: Develop communication, interpersonal and other critical skills which can help them during job.

CO4: Develop professional report writing skill and presentation of their work in industry as well as at institute.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	1	-	-	-	-	-	3	-
CO2	-	-	-	-	-	3	-	3	3	2	-	-	-	-
CO3	-	-	-	-	-	2	-	-	1	3	2	2	-	-
CO4	-	-	-	-	1	-	-	-	1	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME – 344 REFRIGERATION AND AIR-CONDITIONING

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Basic Concepts	2
2.	Refrigerants	2
3.	Air Refrigeration	2
4.	Vapour Compression Refrigeration Cycle	9
5.	Refrigeration System components	3
6.	Vapour Absorption Refrigeration Cycle	6
7.	Psychometry	7
8.	Psychometric Processes	7
9.	Human Comfort	3
10.	Air conditioning systems	4

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|-----|
| 1 | Basic Concepts | 02 Hours | 04% |
| 1.1 | Natural and Mechanical refrigeration, Application of Refrigeration, Units of refrigeration and Coefficient of Performance (COP), Refrigeration effect, cooling capacity and COP of a refrigerator. | | |

1.2	Heating effect, heating capacity and COP as heat pump, Reversed Carnot cycle and its limitations.		
2	Refrigerants	02 Hours	05%
2.1	Classification and nomenclature of refrigerants.		
2.2	Desirable thermodynamic, chemical and physical properties of refrigerants;		
2.3	Comparative study of commonly used refrigerants and their fields of application; Azeotropes, ODP & GWP; Secondary refrigerants; future industrial refrigerants.		
3	Air Refrigeration	02 Hours	05%
3.1	Introduction, Air refrigerator working on Reversed Carnot cycle and Bell Coleman Cycle with their C.O.P of cycles.		
3.2	Numerical of air refrigeration system.		
4	Vapour Compression Refrigeration Cycle	09 Hours	20%
4.1	Vapour compression cycle on P–V, P–H and T–S diagrams, Deviation of actual cycle from theoretical cycle, Analysis of theoretical and actual Vapour compression cycles.		
4.2	Effect of suction pressure, discharge pressure, sub cooling, super heating and pressure drop in valves on performance and cooling capacity.		
5	Refrigeration system components	03 Hours	06%
5.1	Types; construction; working; comparison of Compressors; condensers; expansion devices; and evaporators, evacuation and charging of refrigerant.		
6	Vapour Absorption Refrigeration Cycle	06 Hours	13%
6.1	Principle of absorption system, Simple Vapour absorption refrigeration system, C.O.P of ideal Vapour absorption refrigeration system.		
6.2	Desirable properties of absorption system refrigerants and absorbents.		
6.3	Aqua–ammonia absorption refrigeration system, Lithium bromide–water absorption refrigeration system; comparison between Vapour compression and absorption refrigeration system.		
6.4	Numerical of Vapour absorption refrigeration system.		
7	Psychrometry	07 Hours	16%

- 7.1 Psychrometry properties of air, Dry bulb, Wet bulb and Dew point temperatures; Relative and specific humidity; Degree of saturation; Adiabatic saturation temperature; Enthalpy of air and water vapors; Psychrometric chart.
- 7.2 Numerical on Psychrometry.
- 8 Psychrometric Processes 07 Hours 16%**
- 8.1 Sensible heating and cooling , Latent heating and cooling, Cooling with dehumidification, Cooling with humidification, Heating with dehumidification , Heating with humidification, By-pass factor, Chemical dehumidification, Adiabatic mixing , Air washer.
- 8.2 Numerical of psychrometric processes.
- 9 Human Comfort 03 Hours 06%**
- 9.1 Factors affecting human comfort, effective temperature, factors affecting optimum effective temperature, Basics of Cooling load calculations.
- 10 Air Conditioning Systems 04 Hours 09%**
- 10.1 Air conditioning cycle, classification of air conditioning system, different central air conditioning systems, difference between packaged and central air conditioning plants.
- 10.2 Desert air cooler, window air conditioner, split air conditioner.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Students Learning Outcomes:

On successful completion of the course, the student will be able to:

CO1: Explain the principles and applications of refrigeration systems.

CO2: Determine performance of VCR & VAR system.

CO3: Illustrate desirable properties of refrigerants.

CO4: Describe basics of psychometric and different air conditioning systems.

CO5: Analyze various psychometric processes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	1	-	-	-	-	-	-	-	1	2	-
CO3	2	-	-	-	-	-	1	-	-	-	-	-	-	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	2	-	1	-	-	-	-	-	-	-	1	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Arora C. P, “Refrigeration and Conditioning”, Tata McGraw Hill Publication.
2. Prasad Manohar, “Refrigeration and Conditioning”, Wiley Eastern Limited.
3. Desai P. S., “Modern Refrigeration and air Conditioning for engineers”, Khanna Publications.

Reference Books:

1. Ananthanarayanan, “Refrigeration and Conditioning”, Tata McGraw Hill Publication.
2. Rajput R. K., “Refrigeration and Conditioning”, S. K. Kataria and sons publication.
3. Domkundwar and Arora, “Refrigeration and Air Conditioning”, Dhanpat Rai & Co.
4. Jordan and Priester, “Refrigeration and Conditioning”, Prentice Hall of India.
5. Stoecker W. F., “Refrigeration and Conditioning”, Tata McGraw Hill Publication.

Web Material:

1. <http://nptel.iitm.ac.in/video.php?courseId=1025>

Other material:

1. ASHRAE Hand book (Fundamentals and Equipments)
2. International journal of refrigeration ([www. sciencedirect.com](http://www.sciencedirect.com))

ME – 348 PRODUCTION TECHNOLOGY

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Theory of Metal Cutting	12
2.	Analysis of Machine Tools	04
3.	Semi Automats and Automats	04
4.	Modern Machining Processes	15
5.	Gear and Thread manufacturing	03
6.	Jigs and Fixtures	07

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|---|--------------------------------|-----------------|------------|
| 1 | Theory of Metal Cutting | 12 Hours | 26% |
|---|--------------------------------|-----------------|------------|
- 1.1 Cutting Tool Material, Types of cutting tools, Tool geometry and Force analysis.
 - 1.2 Theory of metal cutting: Orthogonal and oblique cutting, Mechanics of chip formation and types of chips produced, Chip thickness ratio, Shear plane angle and its effect, Forces, Coefficient of friction, Shear strain, Power in machining.
 - 1.3 Merchant circle diagram and its assumptions and use.
 - 1.4 Chip breakers, Tool Dynamometers, Tool wears and methods of tool failure, Tool life.
 - 1.5 Significance of temperature and sources of heat generation, Temperature measurement.
 - 1.6 Cutting fluids and their properties, Economics of machining, Machinability and its evaluation.

- | | | | |
|----------|--|-----------------|-------------|
| 2 | Analysis of Machine Tools | 04 Hours | 09% |
| 2.1 | Working and auxiliary motions in machine tools. Design and analysis of machine tool elements like bed, structure, slide-ways and guide ways and spindle. | | |
| 2.2 | Selection of material, Static and dynamic stiffness, and dynamic rigidity. Drives in machine tools, Stepped regulation of speed and feed, A.P. and G.P. series. | | |
| 3 | Semi Automats and Automats | 04 Hours | 09% |
| 3.1 | Capstan and Turret lathes, Classification of automats, Specifications, Tooling Equipment, Bar stock feeding methods, Universal Chucking Equipment, Tool Layout for Turret, | | |
| 3.2 | Capstan And Automats, Single Spindle And Multi Spindle Automats, Bar Type And Chucking Type Machines Their Principles of Working Constructional Details and Tool Setting. | | |
| 4 | Modern Machining Processes | 15 Hours | 34% |
| 4.1 | Purpose, Need and Classification, Aspects considered in selection of a process. | | |
| 4.2 | Principle, construction, working, process parameters and their influence on machining, selection of parameters and product applications of the following processes: Ultrasonic machining, Abrasive jet machining, Water jet machining. | | |
| 4.3 | Chemical Machining, Electro Chemical Machining and Grinding, Electro discharge Machining, Plasma arc machining, Laser beam machining, Electron beam machining and Hot machining. | | |
| 5 | Gear and Thread Manufacturing | 03 Hours | 07% |
| 5.1 | Gear Manufacturing | | |
| 5.2 | Thread Manufacturing | | |
| 6 | Jigs and Fixtures | 07 Hours | 15 % |
| 6.1 | Usefulness and principles of jig–fixture design, Principles of location. | | |
| 6.2 | Types of locators, Types of clamping devices, Types of bushes, Selection of locators, Clamps and Bushes. | | |
| 6.3 | Types of jigs, Their relative merits, Demerits and Applications, Materials for various elements of jig–fixture, Design of milling, Turning and boring fixtures. | | |

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Students Learning Outcomes:

On successful completion of the course, the student will be able to:

CO1: Identify the types of cutting tools, tool geometry and mechanism of chip removal.

CO2: Analyze cutting forces involved during machining process.

CO3: Identify different machine tool elements and design speed drives in machine tools.

CO4: Categorize semi automats, automats, gear and thread manufacturing methods.

CO5: Execute the applications of various modern machining process and measure material removal rate.

CO6: Associate work holding and tool locating devices for production activities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	1	1	-	1	2	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	1	2	-	-	-	-	-	-	-	-	1	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	1	-	1	-	-	-	-	-	-	-	-	2	-
CO6	1	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Sharma P C, “A Text Book of Production Engineering”, S. Chand Publishers, 11th Edition.
2. P. C. Pandey, “Production Engineering and Science”, Standard Pub., 7th Edition, 2013.
3. HMT, “Production Technology”, Tata McGraw Hill, 2001.

Reference Books:

1. Shaw M. C., “Metal Cutting Practices”, Oxford University Press, 2nd Edition, 2010.
2. Ghosh A. and Mallik A., “Manufacturing Science”, Affiliated East West Pub, 2nd Edition.
3. Mehta N. K., “Machine Tool Design and Numerical Control”, Tata McGraw Hill Publ. Co. Ltd. 3rd Edition.
4. Pandey and Shan, “Modern Machining Processes”, Tata McGraw Hill Publ. Co. Ltd, 2008.
5. Joshi P. H., “Jigs and Fixtures – Design Manual”, Tata McGraw Hill Publ. Co. Ltd., 2nd Edition.
6. Fundamentals of Tool Design – American Society Of Tool Manufacturing, 4th Edition.
7. Donaldson C., “Tool Design”, Tata McGraw Hill, 4th Edition.

Web Material:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html

Other Material:

1. <http://www.sciencedirect.com/science/journal/09240136>
2. <http://www.ieindia.org/publish/pr/pr.htm>
3. <http://www.ieindia.info/public.asp/me/me.htm>
4. <http://www.ias.ac.in/sadhana>

ME354 FLUID MACHINES

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Compressors	12
2.	Impact of Jet	09
3.	Hydraulic Turbine	12
4.	Pumps	12

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Introduction to Compressors 12 Hours 27%
 - 1.1 Reciprocating compressors:
Introduction, Construction and working, actual indicator diagram, Multi-staging, conditions for minimum work, Intercooling, Heat rejected in compressors and intercoolers, mean effective pressure. Indicated power, efficiency of air compressor, merits and demerits of Multi-staging, control of delivery of air compressors, Reciprocating air motors.
 - 1.2 Rotary Compressors:
Introduction, Classification, roots blower, vane type, screw compressor, scroll compressor, centrifugal and axial flow compressors.
 - 1.3 Centrifugal Compressor : Construction and Operation, Ideal energy transfer (Euler's work) velocity diagram, Isentropic efficiency, static and total temperatures, power input factor, slip and slip factor, pressure co-efficient, pre-whirl, effect of blade shape on performance,

losses in centrifugal compressors, blade angles, impeller diameter, impeller width, vane less and vanned diffusers, effect of change in pressure ratio, surging and choking.

- 1.4 Axial flow compressors : Introduction, construction and operation, velocity diagram and work done factor, pressure ratio and static pressure rise, degree of reaction, choice of reaction, blade loading and flow co-efficient, aero foil blading, drag and lift co-efficient. Radial equilibrium theory. Free vortex, forced vortex. Characteristics of curves of centrifugal and axial flow compressors.

2 Impact of Jet 09 Hours 20%

- 2.1 Impact of jet on different types of flat and curved plates
- 2.2 Fixed and moving, Single and series of plates
- 2.3 Derivation of efficiency, Condition for maximum efficiency and value of maximum efficiency, Hinged plate and pipe bends.

3 Hydraulic Turbine 12 Hours 27%

- 3.1 Principles of generation of hydropower, Components of the plant, Layout of plant, Various types of hydro power plants
- 3.2 Introduction and development of turbines, Classification of turbines, Impulse and reaction, Radial and axial, Tangential and mixed flow turbines
- 3.3 Major components of different turbines, Expressions for work done and efficiency of Pelton wheel, Bucket of Pelton wheel, Size and number of buckets, Single jet and Multi jet Pelton wheel
- 3.4 Francis turbine, Kaplan turbine and Propeller turbines, Construction, Working problems on each type
- 3.5 Specific speed, Range of specific speed for different turbines, Performance curves of turbine, Selection of turbines according to available head, Discharge and load, Governing of turbines.

4 Pumps 12 Hours 26%

- 4.1 Classification of pumps, centrifugal pump - velocity diagrams and work done by impeller, losses and efficiencies, pressure rise through impeller, specific speed, NPSH, cavitations, characteristics curves.
- 4.2 Reciprocating Pumps – Operation of single acting and double acting pump, plunger and buckets pumps, work done, slip

4.3 Ideal and actual indicator diagram, effect of velocity and acceleration on suction and delivery pipes, Maximum speed. Air vessels, Miscellaneous pumps like submersible pump, ejector pump etc.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment

D. Students Learning Outcomes:

Upon successful completion of this course, the student will be able to:

CO1: Describe principle, construction and working of various types of Compressors with performance calculations.

CO2: Describe the principal parameters and the action of forces on different geometries by the impact of a fluid jet.

CO3: Predict the complete performance of the turbines theoretically and experimentally.

CO4: Describe the various essential parameters about the system of the hydraulic pumps used in the industry.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	03	02	-	-	-	-	-	-	-	01	-	-	01	-

CO2	02	02	-	-	-	-	-	-	-	-	-	-	01	-
CO3	03	02	-	-	-	-	-	-	-	01	-	-	01	-
CO4	03	02	-	-	-	-	-	-	-	-	-	-	01	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

E. Recommended Study Material:

Text Books:

1. Bansal R. K., "A Text Book of Fluid Mechanics and Hydraulic Machines", Laxmi Publications Pvt. Ltd.
2. Kumar D. S., "Fluid Mechanics and Fluid Power Engineering", S K Kataria and Pub.

Reference Books:

1. Rajput R. K., "Fluid Mechanics and Hydraulic Machines", S Chand and Co. Ltd.
2. Arora K. R., "Fluid Mechanics and Hydraulic Machines", Standard Publishers and Distributors, 2005
3. Jagdish Lal, "Hydraulic", Metropolitan Book Co., 1963
4. Patel V. L. and Patel R. N., "Fluid Power Engineering", Mahajan Publications

Web Material:

1. <http://www.nptel.iitm.ac.in/courses/Webcourse-contents/iitkanpur/machine/ui/toc.html>

Other Material:

- SADHANA (Engineering Science): <http://www.ias.ac.in/sadhana/>
- www.sciencedirect.com: World Pumps, Pump Industry Analysis.
- Institution of Engineers: <http://www.ieindia.info/jourtechdwnlds.aspx>

ME 355 DESIGN OF MACHINE ELEMENTS

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of Hours
1.	Design consideration and Standardization	08
2.	Stress Analysis	08
3.	Power Screw	07
4.	Shaft, Keys, and Couplings	06
5.	Friction Clutches	04
6.	Brakes	04
7.	Cylinders and Pressure Vessel	08

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- 1 Design Considerations and Standardizations 08 Hours 18%**
 - 1.1 Design considerations: Introduction, Design Principles for Manufacture and Assembly, Design principles in Castings, Forgings. Machining, Powder Metallurgy and Welding.
 - 1.2 Standardizations: Introduction, Utility of standardization, Preferred numbers, Standard series and its Applications
 - 1.3 Statistical and Ergonomics Consideration in Design: Reliability based design of machine components and Hazard rate, MTBF
 - 1.4 Aesthetics and Ergonomics in design.

- | | | | |
|----------|---|-----------------|------------|
| 2 | Stress Analysis | 08 Hours | 18% |
| 2.1 | Modes of failure, Theory of Failure under static load, Selection, use and limitation of various failure theories. | | |
| 2.2 | Residual stress, Stress concentration, Causes and Method of Reduction of stress Concentrations. | | |
| 2.3 | Variable stresses, Endurance limits, Factor affecting on endurance limit, Notch Sensitivity. | | |
| 2.4 | Soderberg and Goodman lines for design against fluctuating load. | | |
| 3 | Power Screw | 07 Hours | 16% |
| 3.1 | Design of screw and nut, Design of screw jack and toggle jack. | | |
| 3.2 | Differential and compound screw, Reticulating ball screw. | | |
| 4 | Shaft, Keys, and Couplings | 06 Hours | 13% |
| 4.1 | Shafts: Introduction, Materials used for shaft, Types of shaft, Standard sizes of shaft, Concept of strength and rigidity and Design of shaft | | |
| 4.2 | Keys: Types of keys, Design of square and flat Key, Design of Kennedy key, Design of splines and Effect of keyways | | |
| 4.3 | Couplings: Types of couplings, Design of rigid flange coupling and flexible couplings. | | |
| 5 | Friction Clutches | 04 Hours | 9% |
| 5.1 | Classification, Materials for friction surfaces. | | |
| 5.2 | Consideration in design of friction clutches, Design of single and multi-plate clutches | | |
| 5.3 | Cone clutches, Centrifugal clutches. | | |
| 6 | Brakes | 04 Hours | 9% |
| 6.1 | Introduction, Energy equations. | | |
| 6.2 | Block brakes with short shoe and long shoe, Pivoted block with long shoe. | | |
| 6.3 | Internal expanding brakes, Simple band brake and differential band brake, Principle of disc brakes. | | |
| 7 | Cylinders and Pressure Vessel | 08 Hours | 17% |
| 7.1 | Introduction, Classification. | | |
| 7.2 | Design of thin cylinder and spherical shell subjected internal pressure. | | |

7.3 Design of thick cylinder subjected to internal and external pressure.

7.4 Compound cylinder subjected internal and external pressure.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe the design principles for manufacturing processes and the need of DFMA in product design.

CO2: Explain the need of Standardization and preferred series in design methodology.

CO3: Design and evaluation of fluctuating loading parts subjected to fatigue failure.

CO4: Design machine elements such as Power Screw, Shafts, Keys, Couplings, Brakes, Clutches, and Belt/Chain Drives.

CO5: Design of cylinders and pressure vessels subjected to internal & external pressure.

CO6: Analyze the data and values from design data book to design machine components.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	2	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-	1	-
CO3	3	3	1	-	1	-	-	-	-	-	-	2	3	-
CO4	3	3	2	-	-	-	-	-	1	-	-	3	3	-
CO5	3	3	2	-	-	-	-	-	-	-	-	2	3	-
CO6	2	-	2	-	-	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

1. Bhandari V. B., “Design of Machine Elements”, Tata McGraw Hill Publishing Co.
2. PSG design data book

Reference Books:

1. Haideri Farazadak, “ Machine Design “, Vol.- I, II, III, Nirali Prakashan – Pune.
2. Shigley Joseph Edward, “Machine Design”, McGraw–Hill Professional Publishing.
3. Singh Sadhu, “Machine Design”, Khanna Publishers, New Delhi
4. Sharma P. C. and Aggarwal D. K, “A Text Book of Machine Design: Mechanical Engineering Design”, Katson Publication
5. Dieter George, “Engineering Design”, McGraw–Hill Publishing, 4th Edition 2008.
6. Norton Robert L. “Design of Machinery: An Introduction to the Synthesis and Analysis of Mechanisms and Machines”, Tata McGraw Hill Publishing Co.
7. Sharma C. S. and Purohit Kamlesh, “Design of Machine Elements”, Prentice–Hall of India Pvt. Ltd.– New Delhi
8. Patel R. C. and Pandya A. D., “Machine Design”, Vol.–II, G. Jamnadasand Co.
9. Mott Robert L., “Machine Elements in Mechanical Design”, Prentice Hall, 4th Edition.
10. Juvinall Robert C. and Marshek Kurt M., “Fundamentals of Machine Component Design”, Wiley Publication.

Web material:

1. <http://nptel.iitm.ac.in>
2. Shariff Abdulla, “Hand Book of Properties of Engineering Materials and Materials and Design Data for Machine Elements”.

Other material:

1. Engineering Failure Analysis, Science Direct Journal, ISSN: 1350–6307.
2. Journal of Materials and Design, ELSEVIER PUBLICATION.
3. IE Mechanical Engineering.
4. SADHNA (Engineering Science) (<http://www.ias.ac.in/sadhna/>)

ME 353 COMPUTER AIDED DESIGN

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of CAD	02
2.	Fundamentals of Computer Graphics	12
3.	Geometric Modeling	08
4.	CAD standards	02
5.	Introduction to Finite Element Method	18
6.	Design Optimization	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Fundamentals of CAD 02 Hours 05%
 - 1.1 Introduction to CAD and Design process
 - 1.2 Application of computer for design, Product Cycle and CAD-CAM integration
 - 1.3 Benefits of CAD, CAD work station: Hardware & Software
2. Fundamentals of Computer Graphics 12 Hours 26%
 - 2.1 Introduction to Computer graphics

2.2 Scan conversions and Algorithm for generation of line and circle, Elements of CAD Programming

2.3 2D and 3D Transformations and Composite Transformations

3. Geometric Modeling 10 Hours 18%

3.1 Representation of curves and surfaces

3.2 Geometric modeling techniques

3.3 Wireframe modeling, Surface Modeling and Solid Modeling

3.4 Feature based Parametric modeling

4. CAD standards 03 Hours 05%

4.1 Standards in CAD, Graphics and computing standards

4.2 Data exchange standards, Design database

5. Introduction to Finite Element Method 18 Hours 40%

5.1 Introduction to FEM, Role of FEM in Design cycle and Historical background

5.2 Stresses and equilibrium, Boundary conditions, Strain-Displacement relations, Plane stress and plane strain cases.

5.3 Concept of Raleigh-Ritz and Galerkin's methods, Review of matrix algebra

5.4 Generalized procedure for solving a problem using Finite element analysis

5.5 Types of elements, Finite element modeling and Formulation of Stiffness matrix

5.6 Concept of shape functions and Natural Coordinates

5.7 Thermal effects in 1 D Elements

5.8 1 D structural design problems with Elimination and Penalty Approaches

5.9 Applications and capabilities of various software for FEA

6. Design Optimization 3 Hours 06%

6.1 Introduction to optimum design

6.2 Johnson's method of optimum design

6.3 Normal, Redundant and Incompatible specifications

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory
- Internal Exams/Unit Tests/Surprise Tests/ Quizzes/Seminar/Assignments etc. will be conducted as a part of internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

1. Apply the role of Computer graphics in the geometric modeling.
2. Create a 3D model and technical drawing with a simple, well-defined part or assembly for FE analysis in CAD software
3. Apply parametric modeling techniques to create geometric models.
4. Explain theoretical basis of the Finite Element Method in the engineering design.
5. Explain the need for standardization in computer graphics.
6. Formulate and solve the design problems in mathematical form for optimum design.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	2	-	-	-	-	-	-	2	2	-
CO2	3	3	3	1	3	-			2	1	-	3	3	-
CO3	3	-	-	-	3	-	-	-	-	-	-	3	2	-
CO4	3	3	-	2	3	-	-	-	-	-	-	-	2	-
CO5	2	-	-	-	2	-	-	-	-	-	-	-	1	-
CO6	2	2	-	-	-	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Hearn Donald and Baker Pauline, “Computer Graphics”, Prentice-Hall of India Pvt. Ltd.,
2. Haideri Farazdak, “CAD/CAM and Automation” Rao, Nirali Prakashan.
3. Rao P N, “CAD/CAM: Principles and Applications”, McGraw Hill Publication.
4. Chandrupatla and Belagundu, “Finite Elements in Engineering”, Prentice Hall of India Private Ltd.
5. Logan Daryl, “A first course in the Finite Element Method”, Cengage Learning.

Reference Books:

1. Rogers David F. and Adams J. Alan, “Mathematical Elements for Computer Graphics” McGraw Hill, 1990.
2. Rao S.S., “Finite Element Method in Engineering”, Elsevier Pergaman Press.
3. Lee Kunwoo, “Principles of CAD/CAM/CAE Systems”, Addison Wesley Pub Co.,
4. Zeid Ibrahim, “Mastering CAD/CAM”, Tata McGraw Hill, 2009
5. Chris Mc Mohan, “CAD/CAM: Principles, Practice and Manufacturing”, Prentice Hall.
6. Mortenson M E, “Geometric modeling”, Industrial Press.
7. Reddy J.N., “An Introduction to the Finite Element Method”, McGraw Hill, Int. Edition.
8. Rao S.S., “Engineering optimization: Theory and practice” New Age International Publications.
9. Haideri Farazdak, “Mechanical System Design”, Nirali Prakashan.

Web Material:

1. <http://nptel.ac.in/courses/112102101/>

Other Material:

1. Programming Languages: C, C++, MATLAB
2. Software's: AutoCAD, Pro/Engineer, ANSYS
3. ASME Journal of Mechanical Design (<http://asmedl.aip.org/MechanicalDesign>)
4. IEEE Computer Graphics and Applications
(<https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=38>)
5. IE Mechanical Engg.
6. www.sciencedirect.com
7. SADHNA (Engineering Science) (<http://www.ias.ac.in/sadhana/>)

ME 356 PYTHON FOR MECHANICAL ENGINEERS

6th Semester and 3th Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Basics of Python	02
2.	Numpy Library-working with Arrays/Matrices	02
3.	Plotting the data Matplotlib Library for Python	04
4.	Programs on Mechanics and Machine Design	08
5.	Programs on Mechanical Vibrations	08
6.	Advanced Engineering Programs	06

Total Hours (Lab): 30

Total Hours: 30

B. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Laboratory study involves Python coding for Mechanical Engineering problems.
- Attendance is compulsory in lectures and laboratory.
- Project/Assignments etc. will be conducted as a part of continuous internal evaluation.
- Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Evaluate the given Arrays /Matrices using Numpy Library.

CO2: Develop the Python coding for plotting the data using Matplotlib Library.

CO3: Evaluate given Mechanics and Machine Design problems using Python code.

CO4: Analyze given Mechanical Vibrations/Advanced Mechanical Engineering problems using Python code.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2	-	-	-	-	-	-	-	3	-
CO2	3	3	1	-	-	-	-	-	-	-	-	-	3	-
CO3	3	3	1	-	-	-	-	-	-	-	-	-	3	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Recommended Study Material:

Text Books:

1. Magnus Lie Hetland, "Beginning Python From Novice to Professional", Third Edition, Apress, 2017.
2. Mark Lutz "Learning Python", 4th Edition, O'REILLY, 2016

Reference Books:

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, O'REILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Alex Kenan, "Python for Mechanical and Aerospace Engineering", Alexander Kenan, 2021.

Web Material:

1. <https://www.python.org/>

Software:

1. Python IDLE
2. Anaconda Python

HS132.02 A: CONTRIBUTOR PERSONALITY DEVELOPMENT

6th Semester and 3rd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	–	100	100	

A. Outline of the Course:

Module No.	Title/Topic	Classroom Contact Sessions
1	Concept of Personality: • Meaning of Personality • Types of Personality • Factors contributing to Personality • Personality Traits • Personality Profiling	06
2	Soft Skills and Personality Development: • Positive Thinking and Mind Set • Leadership, Assertiveness and Negotiation Skills • Self-Management • Interpersonal Skills • Being a Team Player	08
3	Developing Contributor Personality • Concept of Contributory Personality • Characteristics of a Contributor • The Contributor's Vision of Success & Career • The Scope of Contribution in a field • Embarking on the Journey to Contributor ship • Developing Contributor Personality • Reviewing Some Contributors Personalities	06

4	Life skills and Personality Development • Concept of life skills • Self-awareness • Empathy • Decision Making • Problem Solving	06
5	Contemporary Issues in CPD • Contemporary Practices & Trends in Contributor Personality Development • Case Study & Presentations	04
Total		30

B. Pedagogy

The course is based on practical learning. Teaching will be facilitated by personality profiling, reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies (Great Personality) , critical reading, group work, independent and collaborative research, presentations etc.

C. Evaluation

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	2	25	
2	Project work	2	25	
3	Personality Profiling and Presentation	1	25	
4	Quiz	2	25	100*
5	Attendance and Classroom participation			05
Total				30

*** Total Marks will be converted to 25**

External Evaluation

The University Exam will be for 70 marks and will be based on practical computer based tests and a viva-voce. It will test the contributory personality aspects and their applications. There may also be a Computer Based Test at the end.

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
Total				70

D. Course Outcomes (COs):

After successfully passing through this course, the students will be able to –

CO1: Identify one's individual personality strengths and challenges.

CO2: Develop more assertive and optimist attitude towards work and life.

CO3: Develop quintessential soft skills to groom one's personality.

CO4: Identify traits of contributor personality.

CO5: Contribute to self, society, nation, and globe.

CO6: Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO3	-	-	-	-	-	1	-	-	3	1	2	1	1	-
CO4	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO5	-	-	2	-	-	2	2	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	2	2	-	-	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Reference Books

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd.

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Electives)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME 380 COMBUSTION PROCESSES

5th Semester and 3rd Year

Program Elective - I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	02
2.	Chemical thermodynamics and combustion chemistry	04
3.	I.C. engine fuels	06
4.	Fuel supply systems & Ignition systems	10
5.	Superchargers	04
6.	Combustion in IC engines	10
7.	Testing of I.C. Engine	06
8.	Pollution by I.C. Engines	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Introduction 02 Hours 4%
 - 1.1 Applications, actual working of IC engines, valve and port timing diagrams.
2. Chemical thermodynamics and combustion chemistry 04 Hours 9%
 - 2.1 Enthalpy of formation, first law applied to reactive system

2.2 Enthalpy and internal energy of combustion, adiabatic saturation temperature.

3. I.C. engine fuels 06 Hours 14%

3.1 Desirable properties of I.C. engine fuels, required qualities of S.I and C.I engine fuels,

3.2 Rating of S.I and C.I. engine fuels, HUOCR, dopes/additives for S.I. and C.I. engines,

3.3 Use of alternate fuels like CNG, LNG, LPG, vegetable oils, biodiesel, alcohol, biogas and hydrogen for IC engines.

4. Fuel supply systems & Ignition systems 10 Hours 22%

4.1 Fuel supply system for SI engines, properties of air-petrol mixture, mixture requirement for different loads and speeds

4.2 Simple carburetor and its working, calculation of air-fuel ratio, types of carburetors, limitations of a single jet carburetor,

4.3 Modern carburetors, problems in carburetors, altitude compensation, gasoline injection in SI engines, mpfi system for modern automobile engines.

4.4 Requirement of ideal injection system, types of injection systems, fuel pumps and injectors,

4.5 Types of nozzles spray formation, quantity of fuel and size of nozzle orifice.

4.6 Ignition system like battery, magneto, and electronic, spark plug, firing order.

5. Superchargers 04 Hours 9%

5.1 Types of superchargers. Supercharging of SI and CI engines,

5.2 effects of supercharging, supercharging limits, methods of supercharging, turbo charging.

6. Combustion in I.C. Engines: 10 Hours 22%

6.1 Stages of combustion, ignition lag and the factors affecting the ignition lag, flame propagation and factors affecting flame propagation

6.2 Abnormal combustion and knocking in SI engines, factors affecting knocking, effects of knocking, control of knocking, combustion chambers for S.I. engines

6.3 Stages of combustion, delay period /ignition lag and the factors affecting it, detonation in C.I. engines,

6.4 Factors affecting detonation, controlling detonation, combustion chambers for C.I. engines.

- | | | |
|---|-----------------|------------|
| 7. Testing of I.C. Engine | 06 Hours | 13% |
| 7.1 Aims of engine testing, measurement of indicated power, brake power, friction power, speed, air consumption, fuel consumption. | | |
| 7.2 IC engine efficiencies, specific output, specific fuel consumption, heat balance sheet, | | |
| 7.3 Performance characteristics of SI and CI engines, testing of IC engines as per Indian standard 10001. | | |
| 8. Pollution by I.C. Engines | 03 Hours | 7% |
| 8.1 Emission of pollutants from SI and CI engines, control of emissions from SI and CI engines, measurement of pollutants in exhaust gases, | | |
| 8.2 Effect of different pollutants on human and plant life , emission (Euro and Bharat stage) norms. | | |

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the actual working of various components of IC engine and the processes.

CO2: Calculate the performance of engine and discuss the effect of various operating variables on engine performance.

CO3: Distinguish normal and abnormal combustion phenomena in SI and CI engines.

CO4: Justify the suitability of conventional and non-conventional fuels for IC engines.

CO5: Summarize the effect of different pollutants on human & plant life and emission

norms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	1	-	-	1	-	-	-	-	-	1	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	1	1	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Domkundwar V. M., “A course in internal combustion engines”, Dhanpatrai and Co. (p) ltd, New Delhi.
2. Mathur and Sharma, “Internal combustion engines“, Dhanpatrai and sons, New Delhi.
3. Ganeshan V, “Internal combustion engines“, Tata Mc Grawhill pub.co. ltd., New Delhi.

Reference Books:

1. Ramalingam, “Internal combustion engines“, Scitech Pub. India Pvt. Ltd., Chennai
2. Gupta H N, “Internal combustion engines“, PHI Learning, New Delhi.

Web Materials:

1. <http://nptel.iitm.ac.in>
2. saeeng.saejournals.org/content/

ME – 373 METAL CASTING TECHNOLOGY

5th Semester and 3rd Year

Program Elective - I

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Metal Casting Technology	02
2.	Metal Casting Science	09
3.	Pattern, Mould and Core Design (Tooling)	06
4.	Feeder and Gating Design (Methoding)	10
5.	Cost Estimation in Casting	04
6.	Design for Castability	04
7.	Advances in Casting Processes	05
8.	Foundry Practices	05

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- 1 Introduction to Metal Casting Technology 02 hours 4%
 - 1.1 Foundry Challenges, Foundry Pressures, Foundry Clusters
 - 1.2 Customers' Requirements , Defining Perfect Castings, Casting Conformance
- 2 Metal Casting Science 09 hours 20%
 - 2.1 Basic physics involved in metal casting process, Metal flow phenomenon: Pressure, velocity and losses, Turbulence and Fluidity
 - 2.2 Solidification phenomena: Liquid solid transformation, nucleation, crystal growth
 - 2.3 Progressive and directional solidification, Analytical solutions of solidification heat transfer, Microstructure and cooling stresses

3	Pattern, Mould and Core Design (Tooling)	06 hours	14%
3.1	Orientation and Parting, Mould Parting Analysis, Pattern Design		
3.2	Cored hole features, Shape Complexity, Core Print Design and Analysis		
3.3	Mould cavity layout		
4	Feeder and Gating Design Analysis (Methoding)	10 hours	22%
4.1	Feeder location and shape, Feeder and Neck design, Methods to achieve directional solidification (Feed aid design)		
4.2	Gating system and types, Gating channel layout, Optimal filling time, Gating element design		
4.3	Computer aided analysis of casting process		
5	Cost Estimation in Metal Casting	04 hours	9%
5.1	Casting process selection, Tooling cost estimation, Material cost estimation, Conversion cost estimation		
6	Design for Castability	04 hours	9%
6.1	Product design for castability, Process friendly design, Castability analysis, Collaborative engineering		
7	Advances in Casting Processes	05 hours	11%
7.1	Near net shape process: Investment Casting & Lost foam casting, VAEPC, Squeeze casting, Thixocasting and Rheocasting, Anitoch plaster mould casting, Slush Casting, Stir casting, Rapid tooling in casting		
8	Foundry Practices	05 hours	11%
8.1	Melting and casting of ferrous and nonferrous alloys, Foundry mechanization, Dust problems in foundry, Foundry environment, health and safety		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Design elements of tooling systems like pattern, mould and core.

CO2: Compute the solidification time in sand and metal die casting.

CO3: Evaluate and design casting feeding and gating system.

CO4: Estimate the costs involved in various stages of metal casting.

CO5: Identify the defects occur in casting and provide the remedial suggestions for elimination.

CO6: Use the CAD software and IOT for foundry applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	-	-	-	-	2	-	-	1	2	-
CO2	3	3	-	2	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	2	-	-	-	-	2	-	-	1	2	-
CO4	2	1	1	-	-	-	-	-	-	-	-	1	-	-
CO5	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO6	1	-	-	-	3	-	2	-	2	2		1	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ravi B., “Metal Casting Computer Aided Design and Analysis”, PHI Learning Pvt. Ltd.
2. Campbell J., “Castings”, 2nd edn, Butterworth-Heinemann, Oxford.

Reference books:

1. Campbell J., “Casting practice: the 10 rules of casting,” Elsevier Butterworth-Heinemann, Oxford
2. Khanna O. P., “Foundry Technology”, Dhanpat Rai Publications
3. Stefanescu D. M., “Science and engineering of casting solidification,” 2nd edn, Springer, New York
4. Beeley P. R., “Foundry Technology”, Butterworth
4. Webster, P. D., “Fundamentals of Foundry Technology”

5. Mukherjee, P. C, “Fundamentals of Metal casting Technology”
6. Bralla J. G., “Design for manufacturability,” 2nd edn, McGraw-Hill, New York

Web materials:

1. <http://efoundry.iitb.ac.in/Academy/index.jsp>
2. <http://nptel.iitm.ac.in/courses.php?branch=Mechanical>

Other materials

1. Indian Foundry Journal
2. International Journal of Metal Casting (AFS-American Foundry Society)
3. AFS Transactions

ME – 379 HYDRAULICS & PNEUMATICS

5th Semester and 3rd Year

Program Elective - I

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No	Title of the Unit	Minimum number of hours
1	Introduction to Fluid power	05
2	Hydraulic Valves and Accessories	07
3	Hydraulic Circuit Design and Analysis	10
4	Basic of Pneumatics Systems	05
5	Pneumatic Actuators and Valves	08
6	Pneumatic circuit design and analysis	10

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

1. Introduction to Fluid power 05 Hours 12%
 - 1.1 Introduction, advantages, disadvantages, application, Principles of Hydraulic Fluid power, fluid properties, Pascal's law, application of Pascal law, continuity equation, hydraulics power, flow rate, , Hydraulic Symbols.
 - 1.2 Introduction to Power pack : Pumping theory, pump classification, gear pump, vane pump, piston pump, pump selection

- 1.3 **Hydraulic actuators and motors:** Introduction, types of cylinder, cylinder force, velocity , power, construction of cylinder, seals in cylinder, mounting style of cylinder, cushioning of cylinder, some common problems in cylinder, introduction to hydraulic motor, gear motor, vane motor, piston motor.
2. **Hydraulic Valves and Accessories:** 07 Hours 15%
- 2.1 **Direction Control Valve:** Construction, classification, valve operating method, center condition of spool valves, Design factors
Pressure Control Valve: Pressure relief valve, pressure reducing valve, unloading valve, sequence valve, Counterbalance valve
Flow control Valve: Throttle valve, throttle with check valve, check valve
- 2.2 **Hydraulics Accessories:** Pressure gauge, pressure intensifier, pressure switch, temperature switch, accumulators, flow meter.
3. **Hydraulic Circuit Design and Analysis:** 10 Hours 22%
- 3.1 Control of single acting and double acting cylinder, meter in and meter out circuits, regenerative circuits, counter valve application, automatic cylinder reciprocating, fail safe circuit, pump unloading circuits, Electro-hydraulic circuits.
4. **Basic of Pneumatics Systems :** 05 Hours 12%
- 4.1 Introduction, application of pneumatics, Gas laws, basic requirements for pneumatic systems
- 4.2 **Air Compressors:** Introduction, types of air compressors, classification. Single acting & double acting cylinders, piston type reciprocating compressor, rotary vane compressor, liquid ring compressor, twin lobe compressor, screw compressor, liquid cooled rotary screw compressor, selection criteria for compressors, characteristics of single acting & double acting cylinder, air receiver
- 4.3 **Servicing Compressed Air:** Introduction, FRL unit, air filter, pressure regulator, lubricator.
5. **Pneumatic Actuators and Valves** 08 Hours 17%
- 5.1 Introduction, types of cylinders, single acting cylinder, double acting cylinder, classification according to construction, cushion assembly, types of mount, piston speed & force, air motors, types of air motors
- 5.2 Types of valves, direction control valves, valve position, basic construction of valves, overlapping,

controlling methods, control techniques, flow control valve, non-return flow control valve, check valve, quick exhaust valve, time delay valve, shuttle valve, twin pressure valve, solenoid valve, cascade systems.

6 Pneumatic circuit design and analysis:

10 Hours 22%

6.1 Symbols, control of single acting & double acting cylinder by various methods, impulse operation, speed control of a cylinder, sequencing of motions, automatic cylinder reciprocating, time delay circuits, logic operation circuits, circuits by cascade systems, Electro-pneumatic circuits

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe the construction, operation principles, and uses of different components of pneumatics and hydraulic system.

CO2: Develop the ability to select correct components of pneumatic system for various applications.

CO3: Read, design, assemble and test basic pneumatic circuits.

CO4: Develop the ability to select correct components of hydraulic system for various applications.

CO5: Read, design, assemble and test basic hydraulic circuits.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-

CO2	2	3	2	-	-	-	-	-	-	-	-	2	3	-
CO3	3	3	3	3	3	-	-	-	-	-	-	2	3	-
CO4	2	3	2	-	-	-	-	-	-	-	-	2	3	-
CO5	3	3	3	3	3	-	-	-	-	-	-	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Majumdar S.R., “Pneumatic systems Principles and maintenance”, Tata McGraw – Hill.
2. Majumdar S.R., “Oil Hydraulics: Principles & Maintenance”, Tata McGraw – Hill
3. John Pippenger & Taylor Hicks, Industrial Hydraulics McGraw-Hill

Reference books:

1. Anthony Esposito, “Fluid Power”, Prentice Hall
2. Andrew Parr, “Hydraulics & Pneumatics” Jaico Publications
3. W. Bolton B/H Butterworth “Pneumatic and Hydraulic system”, Heinemann.

Other Materials:

1. Fundamental of industrial oil hydraulics and Pneumatics
<https://nptel.ac.in/courses/112105046/>

ME – 377 DYNAMICS OF COMPRESSIBLE FLOW

5th Semester and 3th Year

Program Elective- I

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

D. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction	05
2.	Isentropic Flow with Variable Area	08
3.	Normal Shock Waves and Oblique Shock	08
4.	Flow in Constant Area Ducts with Friction	08
5.	Flow in Constant Area Ducts with Heat Transfer	06
6.	Study of Wind Tunnels	04
7.	Boundary Layer Flows	06

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

E. Detailed Syllabus:

1. Introduction 05 Hours 11%
 - 1.1 Thermodynamic concept, Types of flows, Concepts of continuum and control volume.
 - 1.2 Generalized continuity, Momentum and Energy Equations.
 - 1.3 Speed of sound wave, Mach number, Physical difference between incompressible, Subsonic and supersonic flows, three reference speeds, dimensionless velocity, concepts of static and stagnation parameters.
- 2 Isentropic Flow with Variable Area 08 Hours 18%

- 2.1 Concept of Isentropic and adiabatic processes. Mach Number variation, choking in isentropic flow.
- 2.2 Operation of nozzles and diffusers under varying pressure ratios, performance of real nozzles, applications of isentropic flow.
- 3 Normal Shock Waves and Oblique Shock 08 Hours 18%**
- 3.1 Development of a shock wave, governing equations, Rankine – Huguenot, Prandtl and other relations, weak shocks, thickness of shocks, normal shocks in ducts.
- 3.2 Performance of convergent–divergent nozzle with shocks, moving shock waves, shock problems in one dimensional supersonic diffuser, supersonic pitot tube.
- 3.3 Introduction to oblique shock.
- 4 Flow in Constant Area Ducts with Friction 08 Hours 18%**
- 4.1 Fanno curves, Fanno flow equations and its solution, Fanno Flow tables.
- 4.2 Variation of flow properties, performance of long ducts.
- 4.3 Isothermal flow with friction.
- 5 Flow in Constant Area Ducts with Heat Transfer 06 Hours 13%**
- 5.1 Rayleigh line, Rayleigh flow relations, Tables for Rayleigh flow.
- 5.2 Variation of flow properties, choking effects, shock waves with changes in stagnation temperature.
- 6 Study of Wind Tunnels 04 Hours 09%**
- 6.1 Types of wind tunnels
- 6.2 Various flow visualization methods
- 7 Boundary Layer Flows 06 Hours 13%**
- 7.1 Introduction and Description of boundary layer flow, Boundary layer parameters, Boundary layer thickness, Displacement thickness, Momentum thickness, Energy thickness.
- 7.2 Prandtl's boundary layer equation. Characteristic properties of Laminar Boundary Layer. Characteristic properties of Turbulent Boundary Layer.
- 7.3 Separation of boundary Layer. Regimes of external flow-wakes and drag. Drag on immersed body, sphere, cylinder, bluff body, Lift and Magnus effect. Potential flow past a circular cylinder with circulation, doublet. Circulation around aero foil, Drag and Lift of an aero foil.

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe the fundamentals and elements of compressible flows.

CO2: Apply the basic relations in thermodynamics and fluid mechanics required for compressible flow analysis and evaluate the one dimensional steady isentropic compressible flow through variable area duct.

CO3: Analyze one dimensional steady compressible flow through variable area duct with normal shock and oblique shock.

CO4: Evaluate one-dimensional steady compressible flow through a constant area duct with consideration of friction.

CO5: Evaluate one-dimensional steady compressible flow through a constant area duct with consideration of heat transfer.

CO6: Analyze aerodynamics properties of various models with the help of wind tunnel and understand the type of compressible flow by using various flow visualization techniques.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	1	-
CO2	3	3	3	2	1	-	-	-	-	-	-	-	3	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-	3	-
CO4	3	3	3	-	-	-	-	-	-	-	-	-	3	-
CO5	3	3	-	-	-	-	-	-	-	-	-	-	3	-

CO6	2	-		2	1							1	
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1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Yahya S. M., “Fundamentals of Compressible Flow”, New AGE International Limited, 2012.
2. Anderson, J.D., “Modern Compressible Flow: With Historical Perspective”, McGraw Hill, 2003.
3. P. Balachandran, “Fundamentals of Compressible Fluid Dynamics”, PHI Learning Pvt. Ltd., 2006.
4. Kumar, D. S. “Fluid Mechanics and Fluid Power Engineering”, S. K. Kataria & Sons, 2008.

Reference Books:

1. Rathakrishnan. E, “Gas Dynamics”, Prentice, Hall of India Pvt. Ltd., 2008
2. Yahya. S. M., “Gas Tables for Compressible Flow Calculations” New AGE International Limited, 2006.
3. Robert D. Zucker and Oscar Biblarz “Fundamentals of Gas Dynamics” John Wiley & Sons, INC., 2002.
4. Shapiro A. H, “Dynamics and Thermodynamics of Compressible Fluid Flow, Vols. I and II”, Ronald Press, New York, 1953.

Web Material:

1. <http://www.nptel.iitm.ac.in/>

Other Material:

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. www.sciencedirect.com (Experimental Thermal and Fluid Science, International Journal of Heat and Fluid Flow)

ME – 378 NOISE, VIBRATION & HARSHNESS

5th Semester and 3rd Year

Program Elective - I

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of Vibration	02
2.	Two Degree of Freedom system	06
3.	Multi Degree of Freedom system	12
4.	Continuous System and Non Linear Vibrations	09
5.	Noise	06
6.	Noise Generated by Vibrating Structures and Control:	10

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- 1 Fundamentals of Vibration 02 Hours 5%
 - 1.1 Introduction, Single Degree of Freedom Systems
 - 1.2 Free and Forced vibrations, Undamped and Damped Vibrations of Single Degree of Systems

- 2 Two Degree of Freedom system 06Hours 13%
 - 2.1 Undamped Free vibrations of Two degree of freedom system
 - 2.2 Forced Undamped Vibrations of Harmonically Excited Two Degree of Freedom System
 - 2.3 Critical Speed of Shaft and Whirling of Shaft

3	Multi Degree of Freedom system	12Hours	27%
3.1	Matrix iteration method, Stodola Method, Rayleigh Method, Dunkerley's Method, Rayleigh-Ritz method		
3.2	Eigenvalues and Eigenvectors		
3.3	Torsional Vibrations of Three Rotor System		
4	Continuous System and Non Linear Vibrations	09Hours	20%
4.1	Introduction of Continuous Systems		
4.2	Vibration of Strings (Transverse Wave Propagation on a String)		
4.3	Transverse Vibration of Beams		
4.4	Introduction of Non Linear Vibrations,.		
5	Noise	06Hours	13%
5.1	Introduction, Noise Standards and Limits,		
5.2	Vibration As Noise Source, Classification and Analysis Of Machinery Vibrations		
6	Noise Generated by Vibrating Structures and Control	10Hours	22%
6.1	Elementary Noise Radiators; Noise Radiation by Machine;		
6.2	Noise Radiation and Transmission		
6.3	Noise Sources and Identification, Sound Intensity Measurement		
6.4	Control of Noise in Machinery		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Create mathematical model of mechanical system in terms of spring mass and damper.

CO2: Calculate the effect of harmonic force on two degree of freedom system.

CO3: Apply different method of multi degree of freedom system to find out natural frequencies.

CO4: Calculate the natural frequencies of continuous and nonlinear vibrations.

CO5: Illustrate the effect of noise on human comfort and environment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	3	3	-	-	-	-	-	-	-	3	-
CO2	3	3	3	3	2	-	-	-	-	-	-	-	3	-
CO3	3	3	3	3	2	-	-	-	-	-	-	-	3	-
CO4	3	3	3	3	2	-	-	-	-	-	-	-	3	-
CO5	2	2	2	2	1	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Mechanical Vibrations, S. S.Rao, Pearson Education.
2. Mechanical Vibrations, V P Singh, Dhanpat Rai & Co. Ltd
3. Mechanical Vibrations And Noise Engineering, A. G. Ambekar, PHI Learning Pvt Ltd.

Reference books:

1. Mechanical Vibrations S.Graham Kelly and Shashidar K.Kudari, , McGraw-Hill Publishing
2. Fundamentals of Noise and Vibration Analysis for Engineers Norton M P and Karczub D G Cambridge Press.
3. Noise and Vibration Control, IISc Lecture Notes Series: Volume 3,IISc Press, World Scientific.
4. Fundamentals of Noise and Vibration, Frank Fahy and John Walker, CRC Press.

Web Materials

1. <http://www.nptel.iitm.ac.in/>

Other materials

1. Programming Languages & software's: C, C++, MATLAB , CreO
2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>

ME- 38L.01 RENEWABLE ENERGY SOURCES

6th Semester and 3th Year

Program Elective – II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Renewable Energy Sources	2
2.	Solar Thermal Energy	14
3.	Solar Photovoltaic Systems	8
4.	Wind Energy	6
5.	Biogas and Biomass	5
6.	Ocean Energy	5
7.	Geothermal Energy And MHD Power Plants	5

Total Hours (Theory):45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

1	Introduction to Renewable Energy Sources	02	5%
		Hours	
1.1	Man and energy, energy forms, World's and India's production and reserves of energy, Global and national energy scenarios, Need for alternate sources.		
2	Solar Thermal Energy	14 Hours	44%

2.1	Solar geometry, extra-terrestrial radiation, spectral distribution, solar radiation at the earth's surface, earth-sun angles, derived solar angles, sunrise, sunset and day length		
2.2	Instruments for solar radiation measurements, estimation of average solar radiation, radiation on tilted surface. Solar collectors material, types and performance analysis		
2.3	Collector efficiency, overall loss coefficient, collector efficiency factor, solar air heaters- types, performance, applications, focusing collector and its types, tracking, performance, non-focusing type collectors, CPC, optical losses		
2.4	Applications of Solar Energy		
3	Solar Photovoltaic Systems	08	14%
		Hours	
3.1	Introduction, Introduction of semi-conductors and its behaviour, functions of solar cell, Different of solar cell materials, Solar cell performance, Energy losses and solar cell efficiency,		
3.2	Numbers of solar cell required in series and parallel, Photovoltaic panels Application of solar cell systems,		
4	Wind Energy	06	14%
		Hours	
4.1	Introduction, Basics of lift and drag, Energy availability in the wind, power coefficient, Design consideration of wind turbine, Characteristics of windmill rotors, Wind mills- types, Performance, Wind turbine site selection, advantages and disadvantages, applications, wind energy development in India		
4	Biogas and Biomass	06	14%
		Hours	
4.1	Introduction, types of biogas plants, biogas generation, factors affecting biogas generation, design consideration, advantages and disadvantages		
4.2	Applications, biomass energy, energy plantation, gasification, types and application of gasifiers		
5	Ocean Energy	05	11%
		Hours	
5.1	Introduction, OTEC principle, open cycle OTEC system, closed cycle, hybrid cycle		

5.2 Energy from tides, estimation of tidal power, tidal power plants, single basin, double basin, advantages and limitations wave energy, wave energy conversion devices, advantages and disadvantages

6	Geothermal Energy	03	6%
		Hours	

6.1 Introduction, vapour dominated system, liquid dominated system, binary cycle, hot dry rock resources, magma resources, advantages and disadvantages, applications, geothermal energy in India: prospects
Introduction to MHD Power Plants, Principle of MHD power generation, open cycle plant, closed cycle plant, liquid metal system, advantages of MHD plants

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explore the basics about the conventional energy resources and their effective utilization.

CO2: Acquire the knowledge of modern energy conversion technologies.

CO3: Understand principles of energy conversion from alternate sources including solar, wind, geothermal, ocean, biomass and biogas.

CO4: Operate the instrument related to solar radiation measurement and to calculate various solar angles.

CO5: Evaluate performance of solar collectors.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	2	-	-	1	-	-	-	-
CO2	3	2	-	1	-	-	2	-	-	1	-	-	1	-
CO3	3	-	-	-	-	1	2	-	-	1	-	-	-	-
CO4	3	2	-	-	1	1	2	-	-	-	-	1	2	-
CO5	3	2	-	1	1	1	2	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Rai G. D, “Non- Conventional Energy Source”, Khanna Pub.
2. Sukhatme S. P. “Solar Energy”, Tata McGraw Hill Pub

Reference Books:

1. Khan B. H. “Non-conventional energy resources”, Tata McGraw Hill Pub
2. Krieth Frank and Kreider John F, ”Principles of Solar Energy”, John Wiley and sons, New York.
3. Gargand H. P. , Jai Prakash, “Solar Energy : Fundamentals and Applications”, Tata McGraw Hill
4. Duffic J. A. and Beckman W. A. “Solar Engineering of Thermal Processes”, John Wiley and sons, New York.
5. Giri N. K. “Alternate energy sources and application”, Khanna Pub
6. Raja et. al, “Non-conventional energy sources”. Scitech Publications Chennai

Web Materials:

1. <http://nptel.iitm.ac.in/courses.php?disciplineId=112>

Other Materials:

1. www.mnre.gov.in Renewable Energy, Akshay Urja
2. Sadhna: <http://www.ias.ac.in/sadhana/>
3. Sciencedirect.com (Alternative energy sources)

ME – 382.01 INDUSTRIAL ENGINEERING & MANAGEMENT

6th Semester and 3rd Year

Program Elective – II

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	General and organizational Management	05
2.	Production Management	10
3.	Methods of Management	10
4.	Material Management	07
5.	Contemporary issues in Management	13

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

1	General and organizational Management	05 Hours	11%
1.1	Evolution of Management Thoughts		
1.2	Industry 4.0		
1.3	Modern organizational structure.		
2	Production Management	10 Hours	22%
2.1	Production systems		
2.2	Production planning and control		
2.3	Facility planning and layout		

3	Methods of Management	10 Hours	22%
3.1	Method study		
3.2	Time study		
3.3	Ergonomics		
4	Material Management	07 Hours	16%
4.1	Materials management -Concept and classification		
4.2	Inventory Models		
4.3	Purchase management		
4.4	Stores Management		
5	Contemporary issues in Management	13 Hours	29
5.1	Business process Management		
5.2	Performance Management		
5.3	Value Management		
5.4	World Class Manufacturing		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the need of industrial management process in an industry.

CO2: Know the impact of production planning and control activities.

CO3: Identify the principles of motion and time economics.

CO4: Describe the whole process of material managements.

CO5: Analyze the Contemporary issues and their probable solutions with industrial Engineering tools and techniques.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Industrial Engineering and Management NVS Raju Cengage learning Delhi 1st edition 2013
2. A Text Book of Production (operations) Management L. C. Jhamb 7th edition Everest Publishing House, 2012
3. Khanna O. P., “Industrial Engineering and Management”, Dhanpat Rai and sons, New Delhi, 1989
4. Martand Telsang Industrial Engineering and Production Management S Chand & Co Ltd 2006
5. A Text Book of Industrial Engineering L. C. Jhamb 7th edition Everest Publishing House, 2012
6. Industrial Engineering and Production Management M Mahajan 3rd edition 2013
7. “Production and Operations Management”, Pannerselvam Prentice Hall, India, 2nd Edition, 2006.
8. International Labour Organization (ILO), “Introduction to Work Study”, 4th Edition, 2012

Reference Books:

1. Harold B. Maynard., “Maynard's Industrial Engineering Handbook”, McGraw-Hill Education, 2001

2. Parbir Jana ,Manoj Tiwari Industrial Engineering in Apparel Manufacturing Apparel Resources first edition Pvt. Ltd., 2020
3. James A. Tompkins,, “Facilities Planning, Wiley 4th Edition.2010
4. Wallace J. Hopp, Mark L. Spearma, “Factory Physics: 3rd Edition Waveland press 2011.
5. Barnes R.M., “Motion and Time Studies”, John Wiley and Sons, 2004.
6. Stephen N. Chapman Production Planning and Control 1 st edition Pearson 2007
7. Koontz H. and Heinz Weihrich, “Essentials of Management: An International, Innovation, and Leadership Perspective”, McGraw–Hill, education India Pvt ltd New Delhi, 10th Edition.2015

Web Material:

1. <http://nptel.ac.in/courses/Webcourse-contents/IIT-ROORKEE/INDUSTRIAL-ENGINEERING/index.htm>
2. [http://iiie-india.com/\(Indian Institution of Industrial Engineering\)](http://iiie-india.com/(Indian Institution of Industrial Engineering))

Other Material:

1. <http://www.ilo.org/global/lang-en/index.htm>.
2. <https://www.osha.gov>(Journal of Industrial Engineering International)
3. <http://www.springer.com/engineering/industrial+management/journal/40092>
4. <https://www.tandfonline.com/toc/tjci21/current>(Journal of Industrial and Production Engineering)

ME 383.01 DIGITAL MANUFACTURING

6th Semester and 3rd Year

Program Elective -II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Digital Manufacturing	02
2.	Additive Manufacturing	16
3.	Reverse Engineering	07
4.	Computer Aided Quality Control	06
5.	Design aspect for digital manufacturing	06
6.	Virtual Manufacturing	08

Total hours (Theory): 45

Total hours (Lab): 00

Total: 45

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to Digital Manufacturing | 02 hours | 4% |
| 1.1 | Definition of Digital Manufacturing | | |
| 1.2 | Operation Mode and Architecture of Digital Manufacturing System | | |
| 2 | Additive Manufacturing | 16 hours | 35% |
| 2.1 | Introduction to additive manufacturing process, Steps in Additive Manufacturing | | |
| 2.2 | Photo Polymerization Processes | | |
| 2.3 | Powder Bed Fusion Processes | | |
| 2.4 | Printing Processes | | |

2.3	Sheet Lamination Processes		
2.4	Beam Deposition Processes		
2.5	Direct Write Technologies		
2.6	Post processing		
2.7	Challenges, Benefits and Applications		
2.8	Virtual Prototyping		
3	Reverse Engineering	07 hours	15%
3.1	Reverse Engineering– Principles and Technology		
3.2	Reverse engineering hardware and software		
3.3	Applications in Product Manufacturing		
4	Computer Aided Quality Control	06 hours	14%
4.1	Contact and non-contact inspection methods		
4.2	Computer aided testing		
4.3	Machine vision system		
5	Design aspect for digital manufacturing	06 hours	14%
5.1	Parametric modeling		
5.2	Topology aspects		
6	Virtual Manufacturing	08 hours	18%
6.1	Virtual Manufacturing Systems and Environment		
6.2	Augmented Reality Vs Virtual Reality		
6.3	Application of AI and industrial IOT in manufacturing		
6.4	Industry 4.0		
6.5	Data Analytics in Manufacturing		

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the importance of digitalization in today's manufacturing age.

CO2: Use reverse engineering and additive manufacturing in product development.

CO3: Explain application of computer aided testing and inspection in quality control.

CO4: Describe about various design aspects for digital manufacturing.

CO5: Explain the various concepts of Industrial IOT, Industry 4.0, Data analytics, AR vs VR etc. with respect to manufacturing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	1	-	-	-	2	-	-	-	-	-	-	-
CO2	3	2	1	-	3	-	-	-	3	-	-	1	1	-
CO3	1	1	-	-	2	-	-	-	-	-	-	-	1	-
CO4	3	3	2	1	-	-	2	-	-	-	-	-	2	-
CO5	1	-	2	-	2	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Books:

1. Groover M. P., “Automation, Production Systems and Computer Aided Manufacturing”, Pearson Education, New Delhi, (2015)
2. C.K. Chua, K.F. Leong, C.S. Lim, “Rapid Prototyping: Principles And Applications”, 3rd Edition, World Scientific Publishing Co Pte Ltd, (2008)
3. Alasdair Gilchrist, “Industry 4.0: The Industrial Internet of Things”, Apress, (2016)
4. Banerjee P, Zetu D. Virtual manufacturing. John Wiley & Sons; (2001)
5. Zude Z., Shane X., Dejun C. “Fundamentals of Digital Manufacturing Science”, Springer (2012)
6. Ghosh A., “Rapid Prototyping: A Brief Introduction”, East West Press, (2003)
7. Pham D.T. and Dimov S.S, “Rapid Manufacturing”, Springer-Verlag, London, (2001)

8. Raghavendra N. V., Krishnamurthy L., “Engineering Metrology and Measurements”, Oxford University Press (2013)
9. Gibson I, Rosen DW, Stucker B. Additive manufacturing technologies. New York: Springer; (2014)
10. Rayes A, Salam S. Internet of things from hype to reality. The Road to Digitization; River Publisher Series in Communications; Springer: Basel, Switzerland. (2017)
11. Ibrahim Zeid and Sivasubramanian R, “CAD/CAM - Theory and Practice”, Tata McGraw Hill Education, 2011

ME – 384.01 INSTRUMENTATION AND CONTROL

6th Semester and 3rd Year

Program Elective – II

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No	Title of the Unit	Hours
1	Introduction to Mechatronics and Basic Control system	12
2	Sensors and transducers	08
3	Industrial Measurements	10
4	Microprocessor and Micro controller	08
5	Programmable Logic Controller	07

Total hours (Theory): 45

Total hours (Lab): 00

Total: 45

A. Detail Syllabus:

1. Introduction to Mechatronics and Basic Control system 12
Hour 26%
 - 1.1 Introduction to Industry 4.0 Industrial revolution, beginning of industry 4.0, Key components of industry 4.0, Application of industry 4.0
 - 1.2 Introduction to signal conditioning and its necessity , functions of signal conditioning equipment, amplification of signals , types of amplifiers, filters, Introduction to Data Acquisition system, analog to digital conversion, digital to analog conversion

- 1.3 **Basic Control System:** Introduction to control systems, Concept of feedback referred to linear control systems, Definition And Terminology, Mathematical Modeling of Physical Systems, Block Diagram Algebra, Transfer Function From A Block Diagram, Steady – state analysis.
2. **Sensors and transducers** **08 Hour 18%**
- 2.1 **Introduction:** Sensor classification, Sensor characteristics: Transfer function, calibration, hysteresis, non-linearity, repeatability, resolution, dynamic impedance, excitation, dynamic characteristics, reliability, etc.
- 2.2 **Physical Principles of Sensing:** Electric charges, fields and potentials; Capacitance; Magnetism; Induction; Resistance; Piezoelectric effect; pyroelectric effect; Hall effects; Seeback and Peltier effects; Sound waves; Temperature and thermal properties of materials; Heat transfer; Light.
3. **Industrial Measurements** **10 22%
Hour**
- 3.1 **Sensors for displacement** (Potentiometers, Differential transformers, capacitive sensor, Eddy current sensor), Measurement of Acceleration
- 3.2 **Speed Measurement:** Introduction, Mechanical tachometers: Revolution counter, Centrifugal force tachometer, Resonance tachometer Electric tachometers: Eddy current type tachometers, Electric Generator type tachometers, Contactless type tachometers, Frequency type tachometers, Ignition type tachometers, Stroboscopic tachometers, Pneumatic type speed transmitting elements, Measurement of Speed, Frequency and Short Time Intervals by direct application of frequency standards by comparative methods.
4. **Microprocessor and Micro controller** **08 Hour 18%.**
- 4.1 Introduction to analog and digital computers, Difference between an analog and digital computer, Computer peripherals,
- 4.2 Introduction to Microprocessor, Types of microprocessor, Microprocessor system , Features 8085 Microprocessor, Microprocessor programming , Digital logic , Organization of Microcomputers, Types of microcontrollers, introduction of

Microcontrollers, Features of 8051 microcontroller and its architecture, comparison between microprocessor and microcontroller

- 4.3 **Arduino:** Introduction to Arduino, Architecture of Arduino board, Programming in Arduino board, Integration of different sensors with Arduino board

5. Programmable Logic Controller **07 Hour 16%**

- 5.1 Introduction to Programmable Logic Controllers (PLCs), history of PLCs, Advantages of PLCs and Comparison of PLC based control systems with other conventional control systems.
- 5.2 Block diagram of PLC, Internal architecture of PLC, Digital and Analog Input output modules of PLCs, scan cycle and scan time, connections and wiring of various Inputs and Outputs of Plants with the PLC.

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain basic knowledge about mechatronic with data acquisition and Signal conditioning.

CO2: Describe the physical principles used for the sensing elements of different types of sensors.

CO3: Recognize and describe the various sensors used for measurement of typical physical quantities.

CO4: Describe basic concepts of microprocessor, microcontroller and demonstrate the integration of different sensors with Arduino board in real time applications.

CO5: Recognize the use of PLC and their implementation into the industry.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	1	-	-	-	-	-	-	1	-
CO2	3	1	-	-	-	-	-	-	-	-	-	1	2	-
CO3	2	1	-	-	1	-	-	-	-	-	-	-	3	-
CO4	3	3	1	2	-	-	-	-	-	-	-	-	3	-
CO5	2	1	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Dunn C. William, “Fundamentals of Industrial Instrumentation and Process Control”, McGraw-Hill.
2. Rajput R.K., “Mechanical Measurements and Instrumentation”, S. K. Kataria and Sons
3. Rajput R.K., “Mechatronics”, S. Chand Publishing, 2016
4. Jain R. K., “Engineering Metrology”, Khanna Publisher, 2009
Bryan A. L and Bryan A. E., “Programmable Controllers: Theory and Implementation” An Industrial Text Company Publication, 1997

Reference books:

1. Venkateshan S. P., “Mechanical Measurements”, Ane Books India, (2008).
2. Kumar D.S., “Mechanical Measurements and Control”, Metropolitan, New Delhi, 4th Edition
3. W. Bolton, “Instrumentation and control systems”, 2nd edition, Newnes, 2000
5. Bewoor, Vinaykulkarni, “Metrology and Measurement”, Tata McGraw Hill, 1st Edition (2009).
6. Gregory K. McMillan, Process/Industrial Instruments and Controls Handbook, Fifth Edition, McGraw-Hill: New York, 1999.

Other Materials:

1. <https://nptel.ac.in/courses/112103174/3>
2. <https://nptel.ac.in/courses/106105166/>
3. <https://nptel.ac.in/courses/112102011/11>

ME 385.01 AUTOMOBILE ENGINEERING

6th Semester and 3rd Year

Program Elective -II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Automobile & Automobile Performance	02
2.	Chassis, Frame & Body	04
3.	Transmission System	10
4.	Propeller Shaft, Differential & Final Drive	07
5.	Axle, Suspension, Steering and Brakes System	12
6.	Wheels & Tyres	04
7.	Battery, Lighting System, Accessories and Safety	04
8.	Regulation and Standardization of Vehicles	02

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

B. Detailed Syllabus:

- 1 Introduction to Automobile & Automobile Performance 02 Hours 4%
 - 1.1 Development of automobile, classification of automobiles, main parts of automobiles, vehicle assemblies, specifying an automobile
 - 1.2 resistance to the motion of the vehicle,
 - 1.3 Effect of different drives like front wheel/rear wheel/for wheel drive, stability of a vehicle on a slope, dynamics of a vehicle running on banked track, stability of a vehicle taking a turn

- | | | | |
|----------|---|-----------------|------------|
| 2 | Chassis, Frame & Body | 04 Hours | 9% |
| 2.1 | Types of frames, engine location, Comparison of front and rear mounting of engine | | |
| 2.2 | Front and rear differentials, rear, front and four wheel drives, their relative merits | | |
| 2.3 | Types of chassis, pre requirements of body, types of bodies & their construction, aerodynamic considerations in body profiling, agronomical considerations, defects in frames and body | | |
| 3 | Transmission System | 10 Hours | 23% |
| 3.1 | Necessity of a clutch, requirements of a good clutch, used in automobiles | | |
| 3.2 | Constructional features and working of different types of clutch (like single plate/multi plate/cone/ semi centrifugal/ fully centrifugal/wet etc.) | | |
| 3.3 | Nature of wear and tear each components, effect of misalignment and mis- adjustment of components, fluid coupling, trouble shooting in clutch systems | | |
| 3.4 | Functions of gearbox, need of gear box, gears & gear ratios, principle of gearing, | | |
| 3.5 | Types of gear boxes, manual gearboxes, sliding mesh/ constant mesh/ synchromesh type gear box, transfer case of 4 wheel drive vehicle | | |
| 3.6 | Basic devices used in automatic transmission, principle of epicyclic gearing, torque converter, free wheel clutch, over speed drive and its working. | | |
| 3.7 | Semi/fully automatic transmission, continuously variable transmission (CVT) | | |
| 4 | Propeller Shaft, Differential & Final Drive | 07 Hours | 15% |
| 4.1 | Propeller shafts and their types, fluid drive and fluid flywheel, universal joints, Hotchkiss drive, torque tube drive, whirling of propeller shaft | | |
| 4.2 | Principle of the differential, locking differential, limited slip differential | | |
| 4.3 | Final drives and its types, hypoid type final drive | | |
| 5 | Axle, Suspension, Steering and Brake System | 12 Hours | 27% |
| 5.1 | Front Axle: Types, construction, components and their functions, Rear axle, rear axle drives, rear axle shaft supporting, rear axle casing, axle breather, oil retention | | |
| 5.2 | Suspension System: Principle, type of suspension system, conventional and independent front and rear axle, spring, rubber and air suspensions, automatic/hydro suspension system, shock absorbers | | |

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Prepare Vehicle Layouts by understanding Chassis, Frame and Body.

CO2: Interpret the working principle and construction of transmission system and its components.

CO3: Describe design, constructional details and application of automotive suspension and steering systems.

CO4: Describe design, construction details of wheel tyre assembly with automotive braking system.

CO5: Discuss about the automotive electrical and electronics systems.

CO6: Describe procedure of regulation and standardization of vehicles.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	1	-	1	1	-	1	1	1	1	1	1	-
CO2	1	1	2	1	1	1	1	1	1	1	1	1	1	-
CO3	1	1	2	1	1	1	1	1	1	1	1	1	-	-
CO4	1	1	2	1	1	1	1	1	1	1	1	1	-	-
CO5	1	1	2	1	1	1	1	1	1	1	1	1	-	-
CO6	1	1	2	1	1	1	1	1	1	1	1	1		

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Singh Kirpal, “Automobile Engineering” Volume I & II, Standard Pub.& Dist.
2. Gupta K. M. “Automobile Engineering” Volume I & II, Umesh Pub.
3. Gupta R. B. “Automobile Engineering”, Satya Prakashan.
4. Giri N. K., “Automobile Technology”, Khanna Pub.

Reference Books:

1. Crouse W., “Automotive Mechanics”, Tata Mc Graw Hill.
2. Narang G. B. S. “Automobile Engineering”, Khanna Pub.

Web Material:

1. <http://nptel.iitm.ac.in>
2. <http://www.sae.org/pubs/automotive/>
3. www.ijat.net/

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)
2. International Journal of Automotive Technology